

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	20
Flg Calc [Int P]: FLANGE	21
Flg Calc [Int P]: New Flange	29
Internal Pressure Calculations:	35
Element: Channel head- l	36
Element: Channel shell-	38
Element: Shell	40
Element: channel shell-	42
Element: channel head- r	44
Shop Pressure Test Results:	46
Field Pressure Test Results:	48
ASME TS Calc: Pressure Test	50
External Pressure Calculations:	56
Element and Detail Weights:	57
Nozzle Flange MAWP:	60
Wind Load Calculation:	61
Earthquake Load Calculation:	63
Center of Gravity Calculation:	65
Saddle Calcs: Road Transport:	66
Saddle Calcs: Operating Case:	80
Saddle Calcs: Test Case:	95
Saddle Calcs: Ext. Pressure:	109
Nozzle Calcs.: Tube inlet	122
Nozzle Calcs.: Tube outlet	136
Nozzle Calcs.: Shell outlet	150
Nozzle Calcs.: Shell Inlet	164
Nozzle Calcs.: Vent	178
Nozzle Calcs.: Drain	181
ASME TS Calc: DESIGN 1	184
Problems/Failures Summary:	214



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\192-CBEMDia. 1060mm.pvdb

Date of Analysis : Sep 9, 2026 10:11am

PV Elite 28, January 2026



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FileName : 192-C_BEM_Dia

Page 2 of 214

Vessel Design Summary:

Step: 33 10:11am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1060.000 mm ID
Vessel Design Length, Tangent to Tangent 5302.70 mm
Specified Datum Line Distance 50.00 mm
Shell Side Design Temperature 280 °C
Channel Side Design Temperature 275 °C
Shell Side Design Pressure 520.000 KPa.
Channel Side Design Pressure 900.000 KPa.
Shop Shell Side Test Pressure 676.000 KPa.
Field Shell Side Test Pressure 1424.858 KPa.
Shop Channel Side Test Pressure 1170.000 KPa.
Field Channel Side Test Pressure 1326.024 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	2	...	K11572	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head- left	910.420	0.00	2729.000	3.0000	No	No
Channel shell- left	910.420	0.00	2782.000	3.0000	N/A	No
Channel flng- left	900.000	0.00	2315.000	3.0000	N/A	No
Shell	530.420	101.13	2778.000	3.0000	N/A	No
channel flng.- right	900.000	0.00	2315.000	3.0000	N/A	No
channel shell- right	910.420	0.00	2782.000	3.0000	N/A	No
channel head- right	910.420	0.00	2729.000	3.0000	No	No

Liquid Level: 1060.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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FileName : 192-C_BEM_Dia

Page 3 of 214

Vessel Design Summary:

Step: 33 10:11am Sep 9,2026

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.0	16.0	13.6	6.5	...	1.00	1.00
Cylinder	700.0	700.0	14.0	14.0	6.6	...	1.00	1.00
Body Flg	851.3	145.0	108.0	100.0	93.9	93.9	1.00	1.00
Cylinder	4271.4	3340.0	14.0	14.0	5.5	8.6	1.00	1.00
Body Flg	4502.7	145.0	95.0	100.0	93.9	93.9	1.00	1.00
Cylinder	5202.7	700.0	14.0	14.0	6.6	...	1.00	1.00
Ellipse	5252.7	50.0	16.0	13.6	6.5	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	0.964	Shell	Passed
Tube outlet	WRC-297	0.964	Shell	Passed
Shell outlet	WRC-297	0.861	Nozzle	Passed
Shell Inlet	WRC-297	0.812	Nozzle	Passed

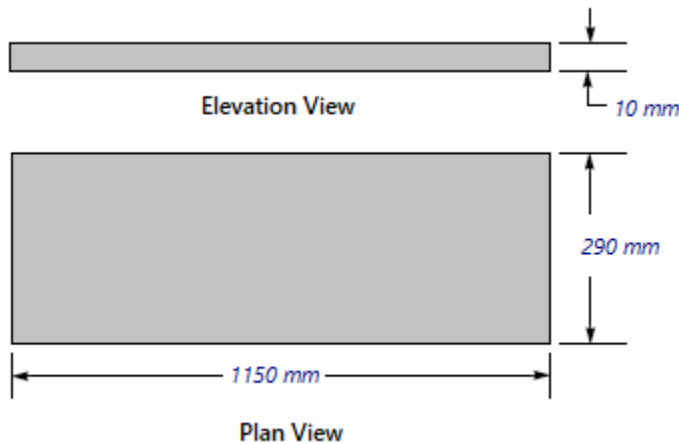
Foundation/Support Summary

Saddle Parameters:

Saddle Width	265.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1000.000 mm
Wear Pad Width	405.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	670.000 mm

Baseplate Length	1150.000 mm
Baseplate Thickness	10.000 mm
Baseplate Width	290.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	470.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch





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FileName : 192-C_BEM_Dia

Page 4 of 214

Vessel Design Summary:

Step: 33 10:11am Sep 9,2026

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	9197.82	Kgf
Maximum Transverse Saddle Shear Load	634.86	Kgf
Maximum Longitudinal Saddle Shear Load	850.27	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	7673.84	Kgf
Maximum Transverse Saddle Shear Load	88.26	Kgf
Maximum Longitudinal Saddle Shear Load	80.61	Kgf

Weights:

Fabricated - Bare w/o removable internals	9482.3	kg.
Shop Test - Fabricated + water (full)	13917.4	kg.
Shipping - Fab. + rem. intls.+ shipping app.	9482.3	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	9882.3	kg.
Empty - Fab. + intls. + details + wghts.	10648.3	kg.
Operating - Empty + operating liquid (no ca)	14818.5	kg.
Field Test - Empty weight + test medium (full)	14228.3	kg.

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Page 5 of 214

Nozzle Summary:

Step: 31 10:11am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube inlet	2572	...	...	OK	6.52	OK	Passed
Tube outlet	2543	...	...	OK	6.56	OK	Passed
Shell outlet	1148	OK	...	OK	5.50	OK	Passed
Shell Inlet	2732	OK	...	OK	5.50	OK	Passed
Vent	2778	...	...	OK	5.50	OK	No Calc[*]
Drain	2778	...	...	OK	5.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1148 Nozzle : Shell outlet [Shellside]
 Minimum MAWP Nozzles : 2543 Nozzle : Tube outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	Tube inlet	400.000	0.000	497.750
20	Tube outlet	400.000	180.000	497.750
40	Shell outlet	3931.350	180.000	206.361
40	Shell Inlet	1331.350	0.000	399.350
40	Vent	2981.350	0.000	88.352
40	Drain	2981.350	180.000	88.352

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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FileName : 192-C_BEM_Dia

Page 6 of 214

Nozzle Schedule:

Step: 30 10:11am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	2.000 in	XXS	LWN	2.375	11.074	...	...	264.86	150
Drain	2.000 in	XXS	LWN	2.375	11.074	...	...	264.86	150
Shell outlet	4.000 in	80	WNF	4.500	8.560	214.30	10.00	267.09	150
Shell Inlet	8.000 in	80	WNF	8.625	12.700	379.08	10.00	275.44	150
Tube inlet	10.000 in	80	WNF	10.750	15.088	433.05	10.00	281.89	150
Tube outlet	10.000 in	80	WNF	10.750	15.088	433.05	10.00	281.89	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-106 B	14.000	20.000	...	...	...
Drain	SA-106 B	11.074	20.000	...	...	...
Shell outle	SA-106 B	14.000	10.000	10.000	10.000	...
Shell Inlet	SA-106 B	14.000	10.000	10.000	10.000	...
Tube inlet	SA-106 B	14.000	15.000	10.000	10.000	...
Tube outlet	SA-106 B	8.000	15.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	2931.350	0.0	250.00	0.00	Shell
Drain	2931.350	180.0	250.00	0.00	Shell
Shell outlet	3881.350	180.0	250.00	0.00	Shell
Shell Inlet	1281.350	0.0	250.00	0.00	Shell
Tube inlet	350.000	0.0	250.00	0.00	Channel shell- 1
Tube outlet	350.000	180.0	250.00	0.00	Channel shell- 1

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FileName : 192-C_BEM_Dia

Page 7 of 214

Bill of Material:

Step: 29 10:11am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 1060.0mm ID X 50.0mm	SA-516 70
1	WELD NECK FLANGE: 108.0mm THK X 1280.0mm OD	SA-266 2
1	WELD NECK FLANGE: 100.0mm THK X 1280.0mm OD	SA-266 2
2	INSULATION: 315mm X 100mm THK	Hot
2	INSULATION: 700mm X 100mm THK	Hot
8	BAFFLES: 1040.0mm DIA X 7.0mm THK	SA-516 70
8	TIE RODS: 25mm DIA X 3000mm	SA-193 B7
2	TUBESHEET: 1280.0mm X 80.0mm THK	SA-266 2
895	TUBES: 3500.0mm X 25.4mm DIA X 2.1mm THK	SA-179
2	SADDLE: 265.0mm X 120 DEG	SA-516 70
1	INSULATION: 3340mm X 100mm THK	Hot
2	GASKET: 1118.0mm OD X 1092.0mm ID	...
72	BODY FLANGE BOLTS: 28.6mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 28.6mm DIA	...
2	CYLINDRICAL SEGMENT 14.0mm THK X 1322.8mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 684.5mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 1285.8mm X 3340.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 1276.5mm X 3340.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 767.8mm X 3340.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 3048.0mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 282.1mm X 700.0mm	SA-516 70
1	NAMEPLATE	...



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FileName : 192-C_BEM_Dia

Page 8 of 214

MDMT Summary:

Step: 32 10:11am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell [8]	B	-18	-48	-29	0.387	14.000	1.00	No
Left Saddle [24]	B	-29	-48	-29	0.387	10.000	1.00	No
New Saddle [24]	B	-29	-48	-29	0.387	10.000	1.00	No
Shell outlet [1]	B	-8	-104	-29	0.052	7.490	1.00	No
Shell Inlet [1]	B	-5	-104	-29	0.054	11.113	1.00	No
Nozzle Flg [5]	A	-18	-104					
Vent [1]	B	-29	-104	-29	0.015	9.690	1.00	No
Nozzle Flg [5]	A	-18	-104					
Drain [1]	B	-29	-104	-29	0.015	9.690	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: SS[13]	B	-7	-104	-29	0.338	20.000	1.00	No
Warmest MDMT:		-5	-48					
Channel flng-[11]	B	-18	-48	-29	0.363	14.000	1.00	No
channel flng.[11]	B	-18	-48	-29	0.363	14.000	1.00	No
Channel head-[10]	B	-19	-48	-29	0.369	13.600	1.00	No
Channel head- [7]	B	-14	-104	-29	0.304	16.000	1.00	No
Channel shell-[8]	B	-18	-48	-29	0.360	14.000	1.00	No
channel shell-[8]	B	-18	-48	-29	0.360	14.000	1.00	No
channel head-[10]	B	-19	-48	-29	0.369	13.600	1.00	No
channel head- [7]	B	-14	-104	-29	0.304	16.000	1.00	No
Tube inlet [1]	B	0	-104		0.093	13.202	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tube outlet [1]	B	-20	-104	-29	0.095	13.202	1.00	No
Nozzle Flg [5]	B	-18	-42					
Tubesheet: CS[14]	B	-7	-104	-29	0.338	20.000	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		0	-42					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-48.0	-12.0	Pass
Channel/Tube	-42.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material



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FileName : 192-C_BEM_Dia

Page 9 of 214

MDMT Summary:

Step: 32 10:11am Sep 9,2026

[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 192-C_BEM_Dia

Page 10 of 214

Warnings and Errors:

Step: 0 10:11am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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FileName : 192-C_BEM_Dia

Page 11 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	520	KPa.
Channel Side Design Pressure	900.02	KPa.
Shell Side Design Temperature	280.0	°C
Channel Side Design Temperature	275.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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FileName : 192-C_BEM_Dia

Page 12 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b)
Pressure Test Position Horizontal
Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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FileName : 192-C_BEM_Dia

Page 13 of 214

Input Echo:

Step: 1 10:11am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Channel head- left
Distance "FROM" to "TO"	50 mm
Inside Diameter	1060 mm
Element Thickness	13.6 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	137 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	210 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation



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FileName : 192-C_BEM_Dia

Page 14 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	-265 mm
Height/Length of Insulation	315 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel shell- left
Distance "FROM" to "TO"	700 mm
Inside Diameter	1060 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube inlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	59.331 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	20
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 15 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	59.331 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-106 B

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Channel flng- left
Distance "FROM" to "TO"	145 mm
Flange Inside Diameter	1060 mm
Element Thickness	100 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	108 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	132.5 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	199 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	3340 mm
Inside Diameter	1060 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	280 °C
Design External Pressure	101.13 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	136.8 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	208.8 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 16 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	670 mm
Width of Saddle	265 mm
Height of Saddle at Bottom	1000 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	405 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	40
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	2670 mm
Width of Saddle	265 mm
Height of Saddle at Bottom	1000 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	405 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3340 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	40
Detail Type	Nozzle
Detail ID	Shell outlet
Dist. from "FROM" Node / Offset dist	2950 mm
Nozzle Diameter	4 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 17 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Weight of Nozzle (Used if > 0)	16.131	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Inlet	
Dist. from "FROM" Node / Offset dist	350	mm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	42.618	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Vent	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.8097	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Drain	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.031	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	670	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	750	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Fireproofing	
Dist. from "FROM" Node / Offset dist	2675	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	750	mm

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Flange
Description	channel flng.- right
Distance "FROM" to "TO"	145 mm



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FileName : 192-C_BEM_Dia

Page 18 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Flange Inside Diameter	1060	mm
Element Thickness	100	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	95	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	900	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	132.5	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	199	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	6/7	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	channel shell- right	
Distance "FROM" to "TO"	700	mm
Inside Diameter	1060	mm
Element Thickness	14	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	900	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	137	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	210	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1060	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	60
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 19 of 214

Input Echo:

Step: 1 10:11am Sep 9,2026

Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	channel head- right
Distance "FROM" to "TO"	50 mm
Inside Diameter	1060 mm
Element Thickness	13.6 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1060 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	315 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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FileName : 192-C_BEM_Dia

Page 20 of 214

XY Coordinate Calculations:

Step: 2 10:11am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Channel head- l		50	...	50	...
Channel shell-		750	...	700	...
Channel flng- l		901.35	...	145	...
Shell		4321.35	...	3340	...
channel flng.-		4552.7	...	145	...
channel shell-		5252.7	...	700	...
channel head- r		5302.7	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 21 of 214

Flg Calc [Int P]: FLANGE

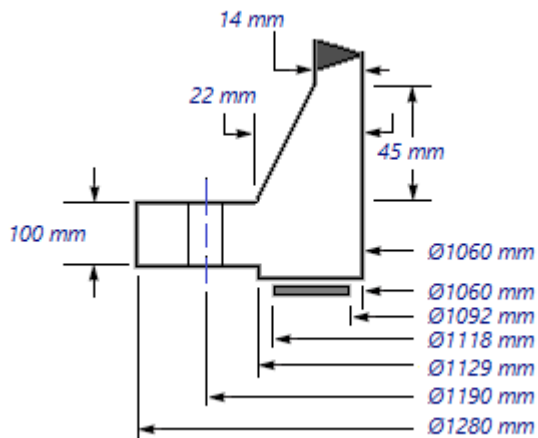
Flng: 5 10:11am Sep 9,2026

Flange Input Data Values

Description: FLANGE :

Channel flng- left

Flange Type	Integral Weld Neck		
Design Pressure	P	900.00	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1060.000	mm
Flange Outside Diameter	A	1280.000	mm
Flange Thickness	t	100.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	22.0000	mm
Length of Hub	h	45.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	132.50	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1190.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1129.000	mm
Flange Face Inside Diameter	Fid	1060.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 22 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (900 \cdot (1060/2 + 3)) / (137 \cdot 1 - 0.6 \cdot 900) + Ca$$

$$= 6.5156 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11)$$

$$= 2792.616 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14)$$

$$= 3588.137 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	97.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1066.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	19.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	43.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1105.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1105^2 \cdot 900$$

$$= 88038.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1105 \cdot 3 \cdot 900$$

$$= 12283.015 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1066^2 \cdot 900$$

$$= 81911.211 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 88038 - 81911$$

$$= 6127.165 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(88038 + 12283 + 0, 0)$$

$$= 100321.391 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 1105 + 0 \cdot 0$$

$$= 156826.797 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((100321 + 0 + 4 \cdot 0/1105)/172, 156827/172)$$

$$= 89.419 \text{ cm}^2$$



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FileName : 192-C_BEM_Dia

Page 23 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am Sep 9,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 97/(3 + 0.5)$$

$$= 223.436 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1190 \cdot \sin(\pi / 36)$$

$$= 103.715 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{104/(2 \cdot 28.6 + 97)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	89.419	169.084	
Radial Distance between Hub and Bolts:	38.100	43.000	
Radial Distance between Bolts and Edge:	28.575	45.000	
Circ. Spacing between the Bolts:	63.500	103.715	223.436

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 169 \cdot 172 / (68.9 \cdot \pi (1118 + 1092))$$

$$= 6.075 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 169$$

$$= 296546.81 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 100321 - 88038$$

$$= 12283.01 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1190 - 1105) / 2$$

$$= 42.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1190 - 1066) / 2 + 42.4)$$

$$= 52.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1190 - 1066 - 19) / 2$$

$$= 52.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 24 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am

Sep 9, 2026

End Pressure, Md	81911.	52.5000	1.0000	4300.
Face Pressure, Mt	6127.	52.2123	1.0000	320.
Gasket Load, Mg	12283.	42.4246	1.0000	521.
Gasket Seating, Matm	296547.	42.4246	1.0000	12581.

Total Moment for Operation, Mo 5141. Kg-m.
Total Moment for Gasket seating, Mg 12581. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 108.287 mm
Hub Ratio, h/h0 = HL / H0 0.416
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.850
Factor V per 4.16.5 0.291
Factor f per 4.16.5 1.137

Stress factor from Table 4.16.4 [K]:

= A / B
= 1280/1066
= 1.201

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.2² • log10(1.2))) / (1.2² - 1))/(1.2 - 1)
= 10.714

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.2²(1 + 8.55246 • log10(1.2)) - 1) / ((1.0472 + 1.9448 • 1.2²) • (1.2 - 1))
= 1.839

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.2²(1 + 8.55246 • log10(1.2)) - 1) / (1.36136(1.2² - 1) • (1.2 - 1))
= 11.773

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.2² + 1) / (1.2² - 1)
= 5.527

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 11.8 • 11² • 108 / 0.29
= 530426.750 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.85/108
= 0.008 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (97 • 0.0079 + 1) / 1.84 + 97³/530427
= 2.679

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.14 • 5141 / 1066) / (2.68 • 19²)
= 55.61 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 25 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am Sep 9,2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.14 \cdot 12581 / 1066) / (2.68 \cdot 19^2)$$

$$= 136.08 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.02 \cdot 5141 / 1066) / (2.68 \cdot 97^2)$$

$$= 3.78 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.02 \cdot 12581 / 1066) / (2.68 \cdot 97^2)$$

$$= 9.25 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (10.7 \cdot 5141 / (97^2 \cdot 1066)) - 5.53 \cdot 3.78$$

$$= 32.95 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (10.7 \cdot 12581 / (97^2 \cdot 1066)) - 5.53 \cdot 9.25$$

$$= 80.64 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (55.6 + \max(3.78, 33)) / 2$$

$$= 44.28 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (136 + \max(9.25, 80.6)) / 2$$

$$= 108.36 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 100321 / 169$$

$$= 58.19 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (156827 / 169)$$

$$= 90.96 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	55.61	198.75	136.08	207.00
Radial Flange	3.78	132.50	9.25	138.00
Tangential Flange	32.95	132.50	80.64	138.00
Maximum Average	44.28	132.50	108.36	138.00
Bolting	58.19	172.00	90.96	172.00

Minimum Required Flange Thickness [Rigidity] 93.904 mm

Estimated M.A.W.P. (Operating) 2315 KPa.

Estimated Finished Weight of Flange at given Thk. 334.6 kg.

Estimated Unfinished Weight of Forging at given Thk 454.4 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 26 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	454.86	454.86	1.000	454.86	499.99	0.910
Tangential Flange	St	267.14	341.14	0.783	267.14	374.99	0.712
Tangential Flange	Sto	170.24	227.43	0.749	170.24	250.00	0.681
Combined	Sh/f + St	663.63	682.29	0.973	663.63	749.99	0.885
Combined	Sr + St	297.79	682.29	0.436	297.79	749.99	0.397
Combined	Sh + Sto	625.10	682.29	0.916	625.10	749.99	0.833
Radial Flange	Sr	30.66	454.86	0.067	30.66	499.99	0.061
Maximum Stress Ratio				1.000			0.910

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.66/1.45)

= 1.1472

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.63/1.45) if $h/h0 < 0.35$, else max (0.25; 0.32/0.75)

= 0.4293

In this case, the hub ratio $h/h0$, 0.4156 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 27 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am

Sep 9, 2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	569.8 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	199.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.478 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1118^2 - 1092^2)$$

$$= 451.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 451 / (36 \cdot 4.7)$$

$$= 555.2 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(555; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 570)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (451 / (36 \cdot 4.7))$$

$$\therefore 507.5 \geq 373.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 451 + \pi/4 \cdot 900 \cdot 1092^2) / (0.7 \cdot 4.7 \cdot 36)$$

$$\therefore 507.5 \geq 441.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 275 (451 / (36 \cdot 4.7))$$

$$\therefore 507.5 \leq 734.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 570 (1/0.48)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 28 of 214

Flg Calc [Int P]: FLANGE

Flng: 5 10:11am Sep 9, 2026

:: 507.5 ≤ 1191.9 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 508 \cdot 0.26 \cdot 4.7 \cdot 28.6$$

$$= 180.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 112 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 12581 / 1 \cdot 9807 \cdot 0.29 / (2.35 \cdot 202713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5141 / 1 \cdot 9807 \cdot 0.29 / (2.35 \cdot 186747 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.443 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 12581 / 1 \cdot 9807 \cdot 0.29 / (2.68 \cdot 202713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.876 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5141 / 1 \cdot 9807 \cdot 0.29 / (2.68 \cdot 186747 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.388 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.36, Temperature Reduction per Fig. UCS 66.1 = 68 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	49.88	207.00	119.01	207.00
Radial Flange	4.41	138.00	10.53	138.00
Tangential Flange	27.06	138.00	64.55	138.00
Maximum Average	38.47	138.00	91.78	138.00
Bolting	58.19	172.00	90.96	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 79.858 mm

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 29 of 214

Flg Calc [Int P]: New Flange

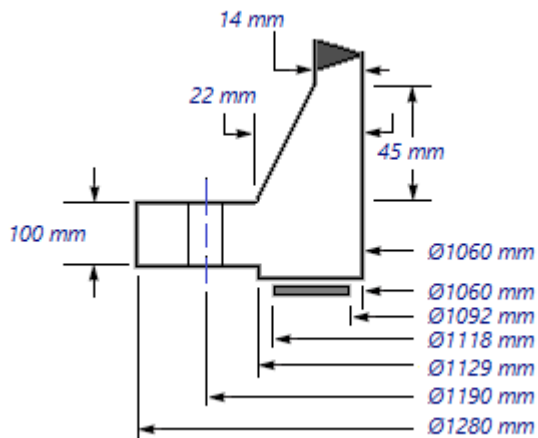
Flng: 6 10:11am Sep 9,2026

Flange Input Data Values

Description: New Flange :

channel flng.- right

Flange Type	Integral Weld Neck		
Design Pressure	P	900.00	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1060.000	mm
Flange Outside Diameter	A	1280.000	mm
Flange Thickness	t	100.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	22.0000	mm
Length of Hub	h	45.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	132.50	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1190.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1129.000	mm
Flange Face Inside Diameter	Fid	1060.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1118.000	mm
Gasket Inside Diameter	Gi	1092.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 30 of 214

Flg Calc [Int P]: New Flange

Flng: 6 10:11am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (900 \cdot (1060/2 + 3)) / (137 \cdot 1 - 0.6 \cdot 900) + Ca$$

$$= 6.5156 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11)$$

$$= 2792.616 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14)$$

$$= 3588.137 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	97.000 mm
Corroded Flange ID, Bcor = B+2*ci	1066.000 mm
Corroded Large Hub, g1Cor = g1-ci	19.000 mm
Corroded Small Hub, g0Cor = go-ci	11.000 mm
Code R Dimension, R = ((C-Bcor)/2) - g1cor	43.000 mm

Gasket Contact Width, N = (Go - Gi)/2	13.000 mm
Basic Gasket Width, bo = N/2	6.500 mm
Effective Gasket Width, b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter, G = Go - 2 * b	1105.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1105^2 \cdot 900$$

$$= 88038.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1105 \cdot 3 \cdot 900$$

$$= 12283.015 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1066^2 \cdot 900$$

$$= 81911.211 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 88038 - 81911$$

$$= 6127.165 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(88038 + 12283 + 0, 0)$$

$$= 100321.391 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 1105 + 0 \cdot 0$$

$$= 156826.797 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((100321 + 0 + 4 \cdot 0 / 1105) / 172, 156827 / 172)$$

$$= 89.419 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 31 of 214

Flg Calc [Int P]: New Flange

Flng: 6 10:11am

Sep 9, 2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 97/(3 + 0.5)$$

$$= 223.436 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1190 \cdot \sin(\pi / 36)$$

$$= 103.715 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{104/(2 \cdot 28.6 + 97)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	89.419	169.084	
Radial Distance between Hub and Bolts:	38.100	43.000	
Radial Distance between Bolts and Edge:	28.575	45.000	
Circ. Spacing between the Bolts:	63.500	103.715	223.436

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 169 \cdot 172 / (68.9 \cdot \pi(1118 + 1092))$$

$$= 6.075 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 169$$

$$= 296546.81 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 100321 - 88038$$

$$= 12283.01 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1190 - 1105) / 2$$

$$= 42.4246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1190 - 1066) / 2 + 42.4)$$

$$= 52.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1190 - 1066 - 19) / 2$$

$$= 52.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 32 of 214

Flg Calc [Int P]: New Flange

Flng: 6 10:11am

Sep 9, 2026

End Pressure, Md	81911.	52.5000	1.0000	4300.
Face Pressure, Mt	6127.	52.2123	1.0000	320.
Gasket Load, Mg	12283.	42.4246	1.0000	521.
Gasket Seating, Matm	296547.	42.4246	1.0000	12581.

Total Moment for Operation, Mo 5141. Kg-m.
Total Moment for Gasket seating, Mg 12581. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 108.287 mm
Hub Ratio, h/h0 = HL / H0 0.416
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.850
Factor V per 4.16.5 0.291
Factor f per 4.16.5 1.137

Stress factor from Table 4.16.4 [K]:

= A / B
= 1280/1066
= 1.201

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.2² • log10(1.2))) / (1.2² - 1))/(1.2 - 1)
= 10.714

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.2²(1 + 8.55246 • log10(1.2)) - 1) / ((1.0472 + 1.9448 • 1.2²) • (1.2 - 1))
= 1.839

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.2²(1 + 8.55246 • log10(1.2)) - 1) / (1.36136(1.2² - 1) • (1.2 - 1))
= 11.773

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.2² + 1) / (1.2² - 1)
= 5.527

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 11.8 • 11² • 108 / 0.29
= 530426.750 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.85/108
= 0.008 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (97 • 0.0079 + 1) / 1.84 + 97³/530427
= 2.679

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.14 • 5141 / 1066) / (2.68 • 19²)
= 55.61 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 33 of 214

Flg Calc [Int P]: New Flange

Flng: 6 10:11am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.14 \cdot 12581 / 1066) / (2.68 \cdot 19^2)$$

$$= 136.08 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.02 \cdot 5141 / 1066) / (2.68 \cdot 97^2)$$

$$= 3.78 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.02 \cdot 12581 / 1066) / (2.68 \cdot 97^2)$$

$$= 9.25 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (10.7 \cdot 5141 / (97^2 \cdot 1066)) - 5.53 \cdot 3.78$$

$$= 32.95 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (10.7 \cdot 12581 / (97^2 \cdot 1066)) - 5.53 \cdot 9.25$$

$$= 80.64 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (55.6 + \max(3.78, 33)) / 2$$

$$= 44.28 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (136 + \max(9.25, 80.6)) / 2$$

$$= 108.36 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 100321 / 169$$

$$= 58.19 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (156827 / 169)$$

$$= 90.96 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	55.61	198.75	136.08	207.00
Radial Flange	3.78	132.50	9.25	138.00
Tangential Flange	32.95	132.50	80.64	138.00
Maximum Average	44.28	132.50	108.36	138.00
Bolting	58.19	172.00	90.96	172.00

Minimum Required Flange Thickness [Rigidity] 93.904 mm

Estimated M.A.W.P. (Operating) 2315 KPa.

Estimated Finished Weight of Flange at given Thk. 334.6 kg.

Estimated Unfinished Weight of Forging at given Thk 454.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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FileName : 192-C_BEM_Dia

Page 34 of 214

Flg Calc [Int P]: New Flange

Flng: 6 10:11am

Sep 9, 2026

$$= 52.14 \cdot 12581 / 1 \cdot 9807 \cdot 0.29 / (2.35 \cdot 202713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5141 / 1 \cdot 9807 \cdot 0.29 / (2.35 \cdot 186747 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.443 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 12581 / 1 \cdot 9807 \cdot 0.29 / (2.68 \cdot 202713 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.876 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5141 / 1 \cdot 9807 \cdot 0.29 / (2.68 \cdot 186747 \cdot 11^2 \cdot 108 \cdot 0.3)$$

$$= 0.388 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.36, Temperature Reduction per Fig. UCS 66.1 = 68 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	49.88	207.00	119.01	207.00
Radial Flange	4.41	138.00	10.53	138.00
Tangential Flange	27.06	138.00	64.55	138.00
Maximum Average	38.47	138.00	91.78	138.00
Bolting	58.19	172.00	90.96	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 79.858 mm

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FileName : 192-C_BEM_Dia

Page 35 of 214

Internal Pressure Calculations: Step: 5 10:11am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Channel head- le		910.42	16	3	1060	137	1
Channel shell- l		910.42	14	3	1060	137	2
Channel flng- le		900	108	3	1060	132.5	3
Shell		530.42	14	3	1060	136.8	4
channel flng.- r		900	95	3	1060	132.5	5
channel shell- r		910.42	14	3	1060	137	6
channel head- ri		910.42	16	3	1060	137	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Channel head- le		900	2729	3532	13.6	6.51822	1
Channel shell- l		900	2782	3588	14	6.55642	2
Channel flng- le		900	2315	2660	100	93.9038	3
Shell		520	2778	3588	14	5.5	4
channel flng.- r		900	2315	2660	100	93.9038	5
channel shell- r		900	2782	3588	14	6.55642	6
channel head- ri		900	2729	3532	13.6	6.51822	7

Warning: Shell Side MAWP governed by a standard Flange!

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1096 KPa.
 Shell Side MAPnc = 1910 KPa.
 Channel Side MAWP = 1020 KPa.
 Channel Side MAPnc = 1580 KPa.

Elements Suitable for Design Internal Pressure.

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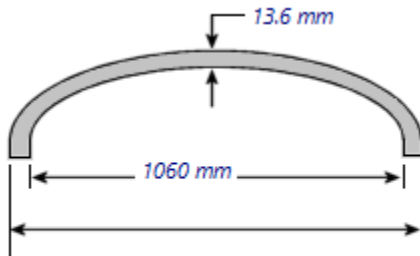
Page 36 of 214

Element: Channel head- 1

Step: 6 10:11am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel head- left

Elliptical Head Design Information:					
Design Pressure	P	910.420 KPa.	Int. Design Temperature		275 °C
Material		SA-516 70	Ext. Design Temperature		25 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1060.000 mm	Actual Stress		45.5 N./mm ²
Spec. Min. Thk t,e, or sa		13.6000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		14641.4 cm ²	SG of Contents		1.000
Element Volume		200063.2 cm ³	Weight of Contents		199.9 Kg
Empty Weight		192.8 Kg	Operating Weight		392.7 Kg
			Comp. All. Ope. Stress		95.9 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (910 \cdot 1066 \cdot 0.99) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 910)$$

$$= 3.5182 + 3.0000 = 6.5182 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.420 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 10.6) / (0.99 \cdot 1066 + 0.2 \cdot 10.6)$$

$$= 2739 - 10.4 = 2729 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 13.6) / (1 \cdot 1060 + 0.2 \cdot 13.6)$$

$$= 3532 \text{ KPa.}$$



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FileName : 192-C_BEM_Dia

Page 37 of 214

Element: Channel head- 1

Step: 6 10:11am

Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (910 \cdot (0.99 \cdot 1066 + 0.2 \cdot 10.6)) / (2 \cdot 1 \cdot 10.6)$$

$$= 45.532 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (910 \cdot 533) / (137 \cdot 1 - 0.6 \cdot 910) + 3$$

$$= 6.556 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 10.420 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 13) / (533 + 0.6 \cdot 13)$$

$$= 3293 - 10.4 = 3283 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1066 / (2 \cdot 268)))^2 / 6$$

$$= 0.992558$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.376 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.6, tr = 3.91, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.37$, Temp. Reduction = 64 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 3.96, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1020.018 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.304	Temperature Difference	90 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm

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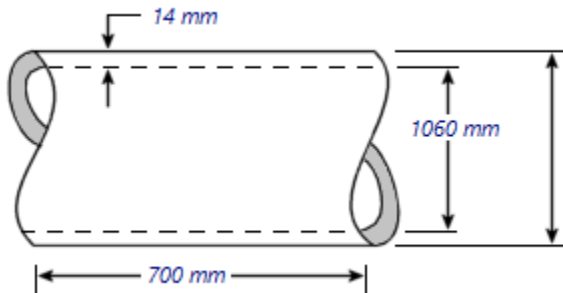
Page 38 of 214

Element: Channel shell-

Step: 7 10:11am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel shell- left

Cylindrical Shell Design Information:			
Design Pressure P	910.420 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1060.000 mm	Actual Stress	44.7 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	23926.4 cm ²	SG of Contents	1.000
Element Volume	617842.1 cm ³	Weight of Contents	617.5 Kg
Empty Weight	256.3 Kg	Operating Weight	873.7 Kg
		Comp. All. Ope. Stress	96.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (910 \cdot 533) / (137 \cdot 1 - 0.6 \cdot 910) \\
 &= 3.5564 + 3.0000 = 6.5564 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.420 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11) \\
 &= 2793 - 10.4 = 2782 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14) \\
 &= 3588 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (910 \cdot (533 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 44.663 \text{ N./mm}^2
 \end{aligned}$$

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FileName : 192-C_BEM_Dia

Page 39 of 214

Element: Channel shell-

Step: 7 10:11am Sep 9, 2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.304 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 3.96$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1020.018 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.360	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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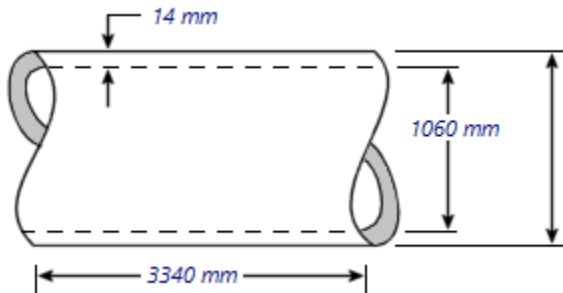
Page 40 of 214

Element: Shell

Step: 8 10:11am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	530.420 KPa.	Int. Design Temperature	280 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	3340.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	136.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1060.000 mm	Actual Stress	26.0 N./mm ²
Spec. Min. Thk t _e , or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	114163.0 cm ²	SG of Contents	1.000
Element Volume	1433017.5 cm ³	Weight of Contents	1432.1 Kg
Empty Weight	1222.8 Kg	Operating Weight	2655.0 Kg
		Comp. All. Ope. Stress	95.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (530 \cdot 533) / (137 \cdot 1 - 0.6 \cdot 530)$$

$$= 2.0716 + 3.0000 = 5.0716 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.420 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11)$$

$$= 2789 - 10.4 = 2778 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14)$$

$$= 3588 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$



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FileName : 192-C_BEM_Dia

Page 41 of 214

Element: Shell

Step: 8 10:11am

Sep 9,2026

$$= (530 \cdot (533 + 0.6 \cdot 11)) / (1 \cdot 11)$$

$$= 26.021 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.304 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 14$, $t_r = 4.25$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.39$, Temp. Reduction = 54 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1096.044 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.387	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1088.00	3340.00	98.91	3.0699	0.0004127	39.95

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 39.9) / (3 \cdot 98.9) = 538 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.595	1088.00	3340.00	194.47	3.0699	0.0001524	14.75

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.8) / (3 \cdot 194) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1088.00	67690.62	98.91	50.0000	0.0001129	10.93

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 10.9) / (3 \cdot 98.9) = 147 \text{ KPa.}$$

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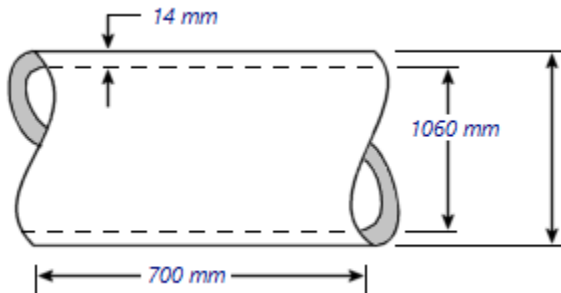
Page 42 of 214

Element: channel shell-

Step: 9 10:11am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: channel shell- right

Cylindrical Shell Design Information:			
Design Pressure P	910.420 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1060.000 mm	Actual Stress	44.7 N./mm ²
Spec. Min. Thk t _e , or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	23926.4 cm ²	SG of Contents	1.000
Element Volume	617842.1 cm ³	Weight of Contents	617.5 Kg
Empty Weight	256.3 Kg	Operating Weight	873.7 Kg
		Comp. All. Ope. Stress	96.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (910 \cdot 533) / (137 \cdot 1 - 0.6 \cdot 910) \\
 &= 3.5564 + 3.0000 = 6.5564 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.420 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137 \cdot 1 \cdot 11) / (533 + 0.6 \cdot 11) \\
 &= 2793 - 10.4 = 2782 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (530 + 0.6 \cdot 14) \\
 &= 3588 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (910 \cdot (533 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 44.663 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 192-C_BEM_Dia

Page 43 of 214

Element: channel shell-

Step: 9 10:11am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.304 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 14$, $t_r = 3.96$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1020.018 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.360	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 44 of 214

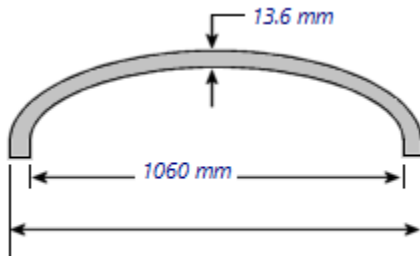
Element: channel head- r

Step: 10 10:11am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: channel head- right

Elliptical Head Design Information:			
Design Pressure P	910.420 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1060.000 mm	Actual Stress	45.5 N./mm ²
Spec. Min. Thk t,e, or sa	13.6000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	14641.4 cm ²	SG of Contents	1.000
Element Volume	200063.2 cm ³	Weight of Contents	199.9 Kg
Empty Weight	192.8 Kg	Operating Weight	392.7 Kg
		Comp. All. Ope. Stress	96.5 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (910 \cdot 1066 \cdot 0.99) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 910)$$

$$= 3.5182 + 3.0000 = 6.5182 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.420 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 10.6) / (0.99 \cdot 1066 + 0.2 \cdot 10.6)$$

$$= 2739 - 10.4 = 2729 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 13.6) / (1 \cdot 1060 + 0.2 \cdot 13.6)$$

$$= 3532 \text{ KPa.}$$



PV Elite 28 Licensee: SL User
 FileName : 192-C_BEM_Dia Page 45 of 214
 Element: channel head- r Step: 10 10:11am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (910 \cdot (0.99 \cdot 1066 + 0.2 \cdot 10.6)) / (2 \cdot 1 \cdot 10.6)$$

$$= 45.532 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (910 \cdot 533) / (137 \cdot 1 - 0.6 \cdot 910) + 3$$

$$= 6.556 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 10.420 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 13) / (533 + 0.6 \cdot 13)$$

$$= 3293 - 10.4 = 3283 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1066 / (2 \cdot 268)))^2 / 6$$

$$= 0.992558$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.376 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.6, tr = 3.91, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.37$, Temp. Reduction = 64 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 3.96, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1020.018 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.304	Temperature Difference	90 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.625 mm



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 46 of 214

Shop Pressure Test Results:

Step: 11 10:11am

Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1424.858 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2483.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1205.649 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1096.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1326.024 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1170.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2054.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1122.020 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1287.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1287.000 KPa.
App 27-4	= M.A.W.P.	1020.018 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 520 • 1

= 676.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 900 • 1

= 1170.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head- left	59.0	234.0	0.252	1180.39
Channel shell- left	57.9	234.0	0.247	1180.39
Shell	33.7	234.0	0.144	686.39
channel shell- right	57.9	234.0	0.247	1180.39
channel head- right	59.0	234.0	0.252	1180.39

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	137.00	1.007
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	137.00	1.007
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	136.80	1.009
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	136.80	1.009
Vent	Nozzle	118.00	118.00	1.000
Drain	Nozzle	118.00	118.00	1.000



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FileName : 192-C_BEM_Dia

Page 47 of 214

Shop Pressure Test Results:

Step: 11 10:11am Sep 9, 2026

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Channel head- left	138.00	137.00	1.007
Channel shell- left	138.00	137.00	1.007
Channel flng- left	138.00	132.50	1.041
Shell	138.00	136.80	1.009
channel flng.- right	138.00	132.50	1.041
channel shell- right	138.00	137.00	1.007
channel head- right	138.00	137.00	1.007

Minimum 1.007

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.15	1.003
Tubesheet Material	138.00	132.50	1.041

Minimum 1.003

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	12.86	216.00	0.060
Tube outlet	12.86	216.00	0.060
Shell outlet	6.78	216.00	0.031
Shell Inlet	7.48	216.00	0.035
Vent	2.29	216.00	0.011
Drain	2.29	216.00	0.011

Elements Suitable for Test Pressure.

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FileName : 192-C_BEM_Dia

Page 48 of 214

Field Pressure Test Results: Step: 12 10:11am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1424.858 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2483.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1205.649 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1096.044 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1326.024 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1170.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2054.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1122.020 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1287.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1287.000 KPa.
App 27-4	= M.A.W.P.	1020.018 KPa.

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head- left	66.8	234.0	0.286	1336.41
Channel shell- left	65.6	234.0	0.280	1336.41
Shell	70.4	234.0	0.301	1435.25
channel shell- right	65.6	234.0	0.280	1336.41
channel head- right	66.8	234.0	0.286	1336.41

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	137.00	1.007
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	137.00	1.007
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	136.80	1.009
Shell Inlet	Nozzle	118.00	118.00	1.000
Shell Inlet	Pad	138.00	136.80	1.009
Vent	Nozzle	118.00	118.00	1.000
Drain	Nozzle	118.00	118.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Channel head- left	138.00	137.00	1.007
Channel shell- left	138.00	137.00	1.007
Channel flng- left	138.00	132.50	1.041
Shell	138.00	136.80	1.009



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FileName : 192-C_BEM_Dia

Page 49 of 214

Field Pressure Test Results: Step: 12 10:11am Sep 9, 2026

channel flng.- right	138.00	132.50	1.041
channel shell- right	138.00	137.00	1.007
channel head- right	138.00	137.00	1.007

Minimum 1.007

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.15	1.003
Tubesheet Material	138.00	132.50	1.041

Minimum 1.003

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	14.56	216.00	0.067
Tube outlet	14.56	216.00	0.067
Shell outlet	14.18	216.00	0.066
Shell Inlet	15.63	216.00	0.072
Vent	4.79	216.00	0.022
Drain	4.79	216.00	0.022

Elements Suitable for Test Pressure.

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FileName : 192-C_BEM_Dia

Page 50 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		1424.86 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1060.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell- left

Channel Type:		Cylinder
Channel Test Pressure		1326.02 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1060.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	895
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3340.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	510.398 mm
Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	40.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm



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FileName : 192-C_BEM_Dia

Page 51 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	80.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1280.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	1129.00 mm
Step 1 Depth on the Tubeside	ht1	6.00 mm
Step 2 Diameter on the Tubeside	dt2	1057.00 mm
Step 2 Depth on the Tubeside	ht2	5.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		1280.00 mm

Note: Tubesheet diameter is now: 1280.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	168.3 cm ²
---------------------------	----	-----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	327.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	1280.000 mm
Flange Inside Diameter	B	1060.000 mm
Flange Face Outside Diameter	Fod	1129.000 mm
Flange Face Inside Diameter	Fid	1060.000 mm
Gasket Outside Diameter	Go	1118.000 mm
Gasket Inside Diameter	Gi	1092.000 mm
Small end Hub thk.	g0	14.0000 mm
Large end Hub thk.	g1	22.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	1190.000 mm
Nominal Bolt Diameter	dB	28.5750 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	36
Bolt Material		SA-193 B7



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 52 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9,2026

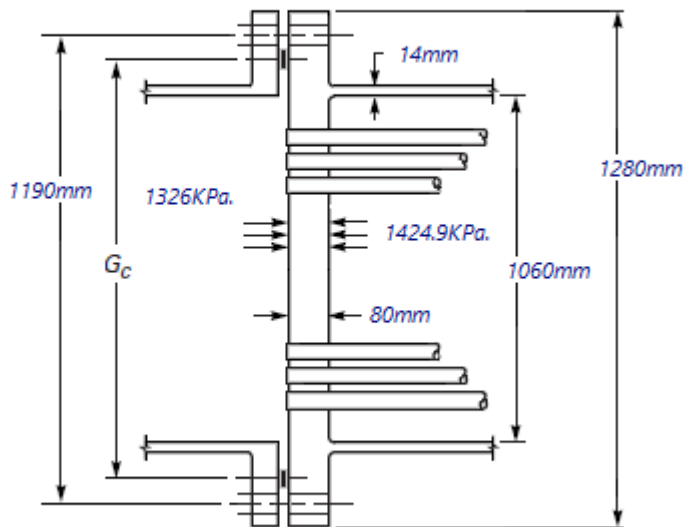
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 44.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 53 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9,2026

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1326 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1326)$

= 0.1812 + 0.0000 = 0.1812 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	3340.00	12.04	50.0000	0.0075908	95.82
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8) / (3 \cdot 12) = 10612$ KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5420	25.40	3340.00	46.86	50.0000	0.0005009	50.08
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 50.1) / (3 \cdot 46.9) = 1425$ KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5420 mm
Allowable Internal Pressure at Corroded thickness	16443.08 KPa.
Required Internal Design Pressure	1326.02 KPa.
Allowable External Pressure at Corroded thickness	10612.12 KPa.
Required External Design Pressure	1424.86 KPa.
Required Thickness due to Shell Side pressure	0.5420 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	46.991	29.885	128	356	5	190	Ps+Pt-Th	D1 Ok
D2uc	18.714	...	30	356	10	190	Ps+Pt-Th	D2 Ok
D3uc	51.630	...	152	356	5	190	Ps+Pt-Th	D3 Ok
Max:	51.6300	29.885 mm		0.426		0.054 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables



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FileName : 192-C_BEM_Dia

Page 54 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9,2026

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	19	370	...	...	:	...	...	...	...	Ok
D2uc	12	370	...	...	:	...	...	...	...	Ok
D3uc	31	370	...	...	:	...	...	...	...	Ok
Max RATIO	0.083				:	...				...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	19	170	-11	-170	292	2691	...	370	...	...	Ok
D2uc	6	170	-6	-170	102	2691	...	370	...	...	Ok
D3uc	12	170	-8	-170	190	2691	...	370	...	...	Ok
Max RATIO	0.109				0.066		0.109		...		...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	51.630	80.000	Ok
Tubesheet thickness flanged extension :	29.885	44.000	Ok
Tube thickness :	0.542	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	3.000	Ok
Tube-tubesheet groove weld leg :	1.333	3.000	Ok

Min Shell length of thk, (14.000) adj. to tubesheet: 219.275 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.035 mm
 Due to Pressure + Thermal Shell Compresses by : -0.035 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0137	...	...	...	...	...



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FileName : 192-C_BEM_Dia

Page 55 of 214

ASME TS Calc: Pressure Test

Case: 1 10:11a Sep 9, 2026

2	-0.0347	...	...	...	...	...
3	-0.0210	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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FileName : 192-C_BEM Dia

Page 56 of 214

External Pressure Calculations: Step: 13 10:11am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	1087.2	10.6	0.0013541	94.0748
20	30	838.333	1088	11	0.0018052	101.388
30	40	No Calc	...	97	No Calc	No Calc
40	50	3340	1088	11	0.00041267	39.9488
50	60	No Calc	...	97	No Calc	No Calc
60	70	838.333	1088	11	0.0018052	101.388
70	80	No Calc	1087.2	10.6	0.0013541	94.0748

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	13.6	No Calc	...	1019
20	30	14	No Calc	...	1367
30	40	100	93.9038	...	No Calc
40	50	14	8.59456	101.13	538
50	60	100	93.9038	...	No Calc
60	70	14	No Calc	...	1367
70	80	13.6	No Calc	...	1019

Minimum

538

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	838.333	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	3340	67690.6	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	838.333	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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FileName : 192-C_BEM_Dia

Page 57 of 214

Element and Detail Weights:

Step: 14 10:11am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	192.796	200063	156.647	203227	19.2796
20	30	256.278	617842	201.924	624856	25.6278
30	40	359.701	136125	347.821	136576	35.9701
40	50	1222.81	1433017	963.465	1466485	122.281
50	60	334.632	128984	323.372	129435	33.4632
60	70	256.278	617842	201.924	624856	25.6278
70	80	192.796	200063	156.647	203227	19.2796
Total		2815	3333936.75	2351	3388662.25	281



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	199.941	-88.3333	...	...	Liquid: 10
10	Insl	71.279	-107.5	...	...	Ins: 40
20	Liqd	617.465	350	...	...	Liquid: 20
20	Insl	92.0694	350	...	...	Ins: 40
20	Nozl	65.2643	350	651.437	...	Tube inlet
20	Nozl	65.2643	350	-651.437	...	Tube outlet
40	Sadl	179.074	670	753	...	Left Saddle
40	Sadl	179.074	2670	753	...	New Saddle
40	Liqd	1432.14	1670	...	...	Liquid: 40
40	Insl	439.303	1670	...	...	Ins: 40
40	Nozl	17.7443	2950	-578.59	...	Shell outlet
40	Nozl	46.8796	350	626.838	...	Shell Inlet
40	Nozl	7.49067	2000	549.088	...	Vent
40	Nozl	9.93409	2000	-549.088	...	Drain
40	Wght	200	670	-750	...	Fire proofing
40	Wght	200	2675	-750	...	Fireproofing
60	Liqd	617.465	350	...	...	Liquid: 60
60	Insl	92.0694	350	...	...	Ins: 60
70	Liqd	199.941	138.333	...	...	Liquid: 70
70	Insl	71.279	157.5	...	...	Ins: 70
30	FTsh	568.343	194.525	...	...	Tubesheet
30	Tube	4123.02	1904.52	...	...	Tubes
30	RTsh	568.343	3614.52	...	...	Tubesheet
40	Bafl	69.2616	327	...	...	Baffles
40	Bafl	69.2616	654	...	...	Baffles
40	Bafl	69.2616	981	...	...	Baffles
40	Bafl	69.2616	1308	...	...	Baffles
40	Bafl	69.2616	1635	...	...	Baffles
40	Bafl	69.2616	1962	...	...	Baffles
40	Bafl	69.2616	2289	...	...	Baffles
40	Bafl	70.2595	2616	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 358.1
Liquid 3067.0



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FileName : 192-C_BEM_Dia

Page 58 of 214

Element and Detail Weights: Step: 14 10:11am Sep 9,2026

Insulation	766.0
Nozzles	212.6
Weights	400.0
Exchanger Components	5814.8
Liquid in Tubes	1103.2

Sum of the Detail Weights 11721.7 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	3096.8	3096.8	3096.8	3096.8	3096.8	3096.8
Saddles	358.1	358.1	358.1	358.1	358.1	358.1
Nozzles	212.6	212.6	212.6	212.6	212.6	212.6
Exchanger	5814.8	5814.8	5814.8	5814.8	5814.8	5814.8
Insulation	...	...	...	...	766.0	766.0
Empty Weights	...	...	...	400.0	400.0	...
Ope Weights	...	...	...	...	...	400.0
Ope. Liquid	...	...	...	...	...	3067.0
Tube Ope Lqd	...	...	...	...	...	1103.2
Test Liquid	...	3331.9	...	...	...	...
Tube Tst Lqd	...	1103.2	...	...	...	...
Totals	9482.3	13917.4	9482.3	9882.3	10648.3	14818.5

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	9482.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	13917.4 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	9482.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	9882.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	10648.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	14818.5 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	14308.6 kg.
Field Test Wt.	- Empty weight + test medium (full)	14228.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1103851.5 cm ³
Weight of Ope Liq in tubes :	1103.2 kg.
Weight of Water in tubes :	1103.2 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	14641.4	13844.7
20	30	23926.4	23310.6
30	40	9946.91	5095.04
40	50	114163	111225
50	60	9625.21	4828.63
60	70	23926.4	23310.6



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FileName : 192-C_BEM_Dia

Page 59 of 214

Element and Detail Weights:

Step: 14 10:11am

Sep 9, 2026

70	80	14641.4	13844.7
----	----	---------	---------

Total	210870.7	195459.2 cm ²
Total	227.0	210.4 ft ²

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FileName : 192-C_BEM_Dia

Page 60 of 214

Nozzle Flange MAWP:

Step: 15 10:11am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Tube inlet	1115.0	1960.0	275	150	GR 1.1	2539.5	-86.4	1115.0	-866.9
Tube outlet	1020.0	1580.0	275	150	GR 2.3	2539.5	-295.4	1020.0	1009.4
Shell outlet	1096.0	1910.0	280	150	GR 2.1	4328.7	1917.4	1096.0	2684.6
Shell Inlet	1096.0	1960.0	280	150	GR 1.1	2935.9	-524.6	1096.0	1291.8
Vent	1096.0	1960.0	280	150	GR 1.1	...	...	...	...
Drain	1096.0	1960.0	280	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1096.044 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1910.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1020.018 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1580.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: None Selected

It appears that the ASME UG-44(b) method of pressure de-rating was not selected. If external loads (forces and bending moments) are also required to be evaluated on standard flanged joints with welding neck flanges as mentioned in UG-44(a)(2), (a)(9) and (a)(10) for vessels constructed per the ASME VIII-1 2019 Code Edition onward, they may be evaluated per the requirements specified in UG-44(b).

You can set the year of construction in Tools->Enter U1 Form Information.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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FileName : 192-C_BEM_Dia

Page 61 of 214

Wind Load Calculation:

Step: 16 10:11am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.92)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.561
Structure Height to Diameter ratio	4.662

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1000 + 300 = 1300 \text{ mm}$$

$$= 4.27 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (4.27 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



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FileName : 192-C_BEM_Dia

Page 62 of 214

Wind Load Calculation:

Step: 16 10:11am Sep 9,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.56 • 4.95

= 37.8 lbs. [17.1] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Channel head- l	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel shell-	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Channel flng- l	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
channel flng.-	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
channel shell-	1300.0	0.000	0.000	0.000	0.849	1.000	78.120
channel head- r	1300.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fire proofing	0.00	300.00	78.12	0.00
Fireproofing	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	1544.64	4599.83	78.12	18.8495
20	30	1300	1545.6	10819.2	78.12	44.3357
30	40	1300	1272	1844.4	78.12	7.55812
40	50	1300	1545.6	51623	78.12	211.545
50	60	1300	1272	1844.4	78.12	7.55812
60	70	1300	1545.6	10819.2	78.12	44.3357
70	80	1300	1544.64	4599.83	78.12	18.8495

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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FileName : 192-C_BEM_Dia

Page 63 of 214

Earthquake Load Calculation:

Step: 17 10:11am

Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(5.020.75)$
 $= 0.067$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.067, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 14460 \cdot 1.25$
 $= 850.267$ Kg

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 14460$
 $= 453.5$ Kg

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kg	Element Ope Load Kg
10	20	530	1606.7	94.4741



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FileName : 192-C_BEM_Dia

Page 64 of 214

Earthquake Load Calculation:

Step: 17 10:11am

Sep 9, 2026

20	30	530	1606.7	94.4741
30	40	530	1606.7	94.4741
40	Sad1	530	1606.7	94.4741
Sad1	50	530	1606.7	94.4741
40	50	530	1606.7	94.4741
50	60	530	1606.7	94.4741
60	70	530	1606.7	94.4741
70	80	530	1606.7	94.4741

The Earthquake Weight does not include the weight of the saddles.

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FileName : 192-C_BEM_Dia

Page 65 of 214

Center of Gravity Calculation: Step: 18 10:11am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2651.350 mm
Center of Gravity of the Liquid	2651.350 mm
Center of Gravity of the Insulation	2651.350 mm
Center of Gravity of the Nozzles	1111.751 mm
Center of Gravity of the Added Weights (Operating)	2653.850 mm
Center of Gravity of the Added Weights (Empty)	2653.850 mm
Center of Gravity of the Tubesheet(s)	2654.525 mm
Center of Gravity of the Tubes	2654.525 mm
Center of Gravity of the Baffles	2454.907 mm
Center of Gravity of Bare Shell New and Cold	2635.121 mm
Center of Gravity of Bare Shell Corroded	2632.404 mm
Vessel CG in the Operating Condition	2618.912 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	2603.903 mm
Vessel CG in the Test Condition	2617.937 mm

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FileName : 192-C_BEM_Dia

Page 66 of 214

Saddle Calcs: Road Transport:

Step: 19 10:11am

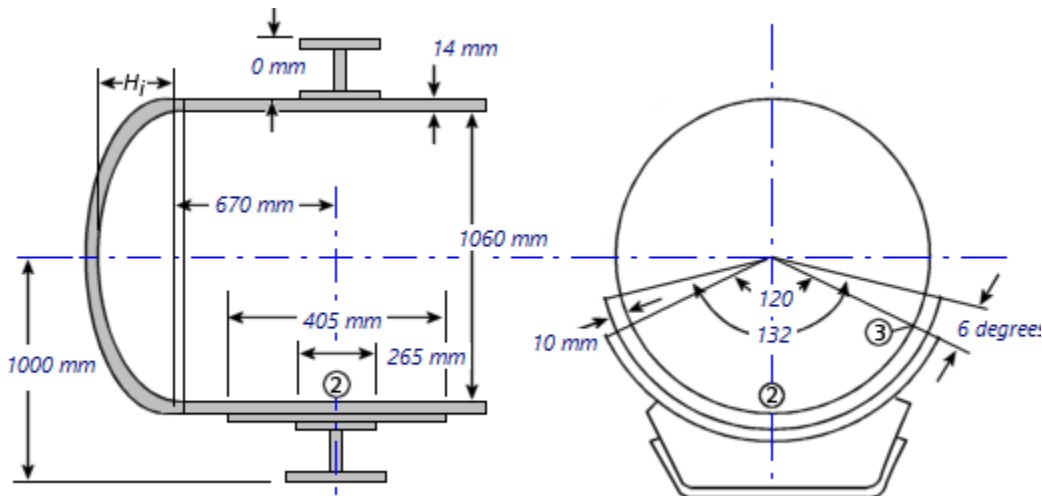
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	537.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		5907.98	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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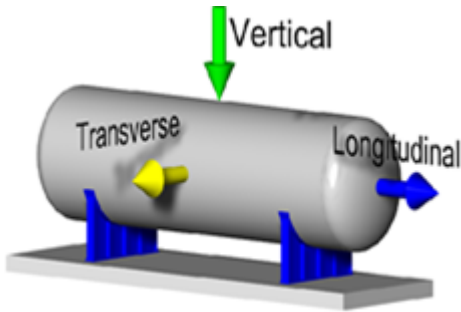
FileName : 192-C_BEM_Dia

Page 67 of 214

Saddle Calcs: Road Transport:

Step: 19 10:11am

Sep 9, 2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.95	-234.00
Long. Stress at Top of Midspan	0.95	136.80
Long. Stress at Bottom of Midspan	0.95	136.80
Long. Stress at Top of Saddles	3.91	136.80
Long. Stress at Bottom of Saddles	-2.17	-234.00
Long. Stress at Bottom of Saddles	2.17	136.80
Tangential Shear in Shell	5.40	109.44
Circ. Stress at Horn of Saddle	11.76	205.20
Circ. Compressive Stress in Shell	0.46	136.80

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 92.7 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 787.7 \text{ lbf} \\
 &= 357.292 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 4.23^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 143.0 \text{ lbf} \\
 &= 64.8 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 9124 \cdot 1 + 357 \\
 &= 1269.7 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 9124 \cdot 1 + 64.8 \\
 &= 977.3 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 1270/2 \cdot 1000/960 \\
 &= 661.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 977 \cdot 1000/2000 \\
 &= 488.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 4997 \cdot 0.05
 \end{aligned}$$



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FileName : 192-C_BEM_Dia

Page 68 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9, 2026

= 249.827 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 4997 + 250 + max(489, 662)

= 5908.0 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6087.05	Kgf
Transverse Shear Load	634.86	Kgf
Longitudinal Shear Load	977.26	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(1060/2 + 14 + 10)sin(120/2), 2•537)

= 959.556 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -5908•670[1-(1-670/3340+(537²-0²)/

(2•670•3340))/(1+(4•0)/(3•3340))]

= -539.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 5908•3340/4((1+2(537²-0²)/(3340²))/(1+(4•0)/

(3•3340))-4•670/3340)

= 1229.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 537/(2•14) - 1230/(π•537²•14)

= -0.95 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 537/(2 • 14) + 1230/(π • 537² • 14)

= 0.95 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•537/(2•14)--539/(0.11•π•537²•14)

= 3.91 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•537/(2•14)+-539/(0.19•π•537²•14)

= -2.17 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 5908 (3340 - 2 • 670)/(3340 + (4 • 0/3))

= 3537.7 Kgf



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FileName : 192-C_BEM_Dia

Page 69 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 3538 / (537 \cdot 14)$$

$$= 5.40 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(537 \cdot 14)}$$

$$= 67.631 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{(537 \cdot 14)}, 2 \cdot 670)$$

$$= 400.26 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 5908 \cdot 0.1 / (400(14 + 1 \cdot 10))$$

$$= -0.46 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -5908 / (4(14 + 1 \cdot 10)400) -$$

$$12 \cdot 0.053 \cdot 5908 \cdot 537 / (3340(14 + 1 \cdot 10)^2)$$

$$= -11.76 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(530 \cdot 14)} + 405)14^3) / 12$$

$$= 12.334 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 12.3 + 75.5(134 - 7)^2$$

$$= 12253.053 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	539.4	14.0	7.0	75.5	52859.0	0.123E+05
Wearplate	405.0	10.0	19.0	40.5	76950.0	0.539E+04
Web	10.0	436.0	242.0	43.6	1055119.9	0.120E+05
BasePlate	290.0	10.0	465.0	29.0	1348499.8	0.317E+05
Totals	...	...	...	188.6	2533428.8	0.613E+05

Distance to Centroid [C1]:

$$= AY / A$$



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FileName : 192-C_BEM_Dia

Page 70 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

$$= 2533428/189$$

$$= 134.319 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 5908$$

$$= 1202.4020 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.0426 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 530 \cdot \sin(120/2) / 1.05$$

$$= 561.694 \text{ mm}$$

Bending Moment, M = Fh * d = 675.3947 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.4509 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5908 + 179)290 / (4 \cdot 1150 \cdot 173))}$$

$$= 8.071 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10)$$

$$= 959.556 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 960 / (4 - 1)$$

$$= 319.852 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 320 \cdot 290 / 2$$

$$= 463.786 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6087 / (1150 \cdot 290)$$

$$= 1.825 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 464 \cdot 1.83$$

$$= 846.503 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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FileName : 192-C_BEM_Dia

Page 71 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

$$= 25.5 + 16$$

$$= 41.493 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 847/41.5$$

$$= 2.001 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 265 - 10$$

$$= 255.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/41.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1551/41.5)}$$

$$= 61.135 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (530 + 14 + 10) - 10$$

$$= 713.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 713/61.1$$

$$= 11.663$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 977 / (2 \cdot 1150) \cdot 320 \cdot 713/2$$

$$= 48.451 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2/152 + (48.5 \cdot 132/15508023) / 173$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 72 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9, 2026

= 0.037

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/57.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 265/2

= 132.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1000 - $\sqrt{(554 + 10)^2 - (320/2)^2}$ - 10

= 459.585 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(1553/57.5)}$

= 51.983 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 460/52

= 8.841

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 977/(2 • 1150)

= 0.425 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.42 • 320 • 460

= 62.461 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 62.5/117237

= 5.225 N./mm²

Compressive Allowable, KL/r < Cc (8.84 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (8.84)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84^3) / (8 \cdot 123^3))$

= 153.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 2.89/153 + (62.5 • 132/1553)/173

= 0.049

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 73 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

Vessel Mean Radius	Rm	537.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		5415.65	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-0.87	-234.00
Long. Stress at Top of Midspan		0.87	136.80
Long. Stress at Bottom of Midspan		0.87	136.80
Long. Stress at Top of Saddles		3.58	136.80
Long. Stress at Bottom of Saddles		-1.99	-234.00
Long. Stress at Bottom of Saddles		1.99	136.80
Tangential Shear in Shell		4.95	109.44
Circ. Stress at Horn of Saddle		10.78	205.20
Circ. Compressive Stress in Shell		0.42	136.80

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 92.7 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 787.7 \text{ lbf} \\
 &= 357.292 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 4.23^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 143.0 \text{ lbf} \\
 &= 64.8 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 9124 \cdot 1 + 357
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 74 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9, 2026

= 1269.7 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 9124 \cdot 1 + 64.8$

= 977.3 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $1270/2 \cdot 1000/960$

= 661.6 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $977 \cdot 1000/2000$

= 488.6 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $4528 \cdot 0.05$

= 226.383 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $4528 + 226 + \max(489, 662)$

= 5415.6 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

5594.72 Kgf

Transverse Shear Load

634.86 Kgf

Longitudinal Shear Load

977.26 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(1060/2 + 14 + 10) \sin(120/2), 2 \cdot 537)$

= 959.556 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-5416 \cdot 670 [1 - (1 - 670/3340 + (537^2 - 0^2)/$

$(2 \cdot 670 \cdot 3340) / (1 + (4 \cdot 0) / (3 \cdot 3340))]$

= -494.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $5416 \cdot 3340/4 ((1 + 2(537^2 - 0^2)/(3340^2)) / (1 + (4 \cdot 0)/$

$(3 \cdot 3340)) - 4 \cdot 670/3340)$

= 1127.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$

= $0 \cdot 537 / (2 \cdot 14) - 1127 / (\pi \cdot 537^2 \cdot 14)$

= -0.87 N./mm²



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 75 of 214

Saddle Calcs: Road Transport:

Step: 19 10:11am

Sep 9, 2026

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 537 / (2 \cdot 14) + 1127 / (\pi \cdot 537^2 \cdot 14)$$

$$= 0.87 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 537 / (2 \cdot 14) - 494 / (0.11 \cdot \pi \cdot 537^2 \cdot 14)$$

$$= 3.58 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 537 / (2 \cdot 14) + 494 / (0.19 \cdot \pi \cdot 537^2 \cdot 14)$$

$$= -1.99 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 5416 \cdot (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0 / 3))$$

$$= 3242.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 3243 / (537 \cdot 14)$$

$$= 4.95 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{537 \cdot 14}$$

$$= 67.631 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{537 \cdot 14}, 2 \cdot 670)$$

$$= 400.26 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 5416 \cdot 0.1 / (400(14 + 1 \cdot 10))$$

$$= -0.42 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -5416 / (4(14 + 1 \cdot 10)400) -$$

$$12 \cdot 0.053 \cdot 5416 \cdot 537 / (3340(14 + 1 \cdot 10)^2)$$

$$= -10.78 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	470.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 76 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(530 \cdot 14)} + 405)^3) / 12$$

$$= 12.334 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 12.3 + 75.5 (134 - 7)^2$$

$$= 12253.053 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	539.4	14.0	7.0	75.5	52859.0	0.123E+05
Wearplate	405.0	10.0	19.0	40.5	76950.0	0.539E+04
Web	10.0	436.0	242.0	43.6	1055119.9	0.120E+05
BasePlate	290.0	10.0	465.0	29.0	1348499.8	0.317E+05
Totals	...	...	...	188.6	2533428.8	0.613E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2533428 / 189$$

$$= 134.319 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 5416$$

$$= 1102.2021 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.9557 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1000 - 530 \cdot \sin(120 / 2) / 1.05$$

$$= 561.694 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 619.1121 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.3299 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5416 + 179) \cdot 290) / (4 \cdot 1150 \cdot 173)}$$

$$= 7.737 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (530 + 14 + 10)$$

$$= 959.556 \text{ mm}$$

Distance between Ribs [e]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 77 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

= Web Length / (Nribs - 1)
 = 960/(4 - 1)
 = 319.852 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 320 • 290/2
 = 463.786 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 5595/(1150 • 290)
 = 1.678 Kg/cm²

Axial Load [P]:

= Ap • Bp
 = 464 • 1.68
 = 778.037 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 25.5 + 16
 = 41.493 cm²

Compressive Stress [Sc]:

= P/Ar
 = 778/41.5
 = 1.839 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 265 - 10
 = 255.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/41.5
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 265/2
 = 132.500 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)
 = √(1551/41.5)
 = 61.135 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1000 - cos(120/2)(530 + 14 + 10) - 10
 = 713.000 mm

Intermediate Term [Cc]:

= √(2 • π² • Elastic Modulus / Yield Stress)
 = √(2 • π² • 199943008/260)



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 78 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9,2026

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 713/61.1$$

$$= 11.663$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 977 / (2 \cdot 1150) \cdot 320 \cdot 713/2$$

$$= 48.451 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.84/152 + (48.5 \cdot 132/15508023) / 173$$

$$= 0.036$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/57.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1000 - \sqrt{ (554 + 10)^2 - (320/2)^2 } - 10$$

$$= 459.585 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1553/57.5 }$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 977 / (2 \cdot 1150)$$

$$= 0.425 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.42 \cdot 320 \cdot 460$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 79 of 214

Saddle Calcs: Road Transport: Step: 19 10:11am Sep 9, 2026

= 62.461 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 62.5/117237

= 5.225 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.84 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(8.84)²/(2 • 123²))260/

(5/3+3•(8.84)/(8• 123)-(8.84³)/(8•123³)

= 153.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 2.66/153 + (62.5 • 132/1553)/173

= 0.047

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 192-C_BEM_Dia

Page 80 of 214

Saddle Calcs: Operating Case:

Step: 20 10:11am

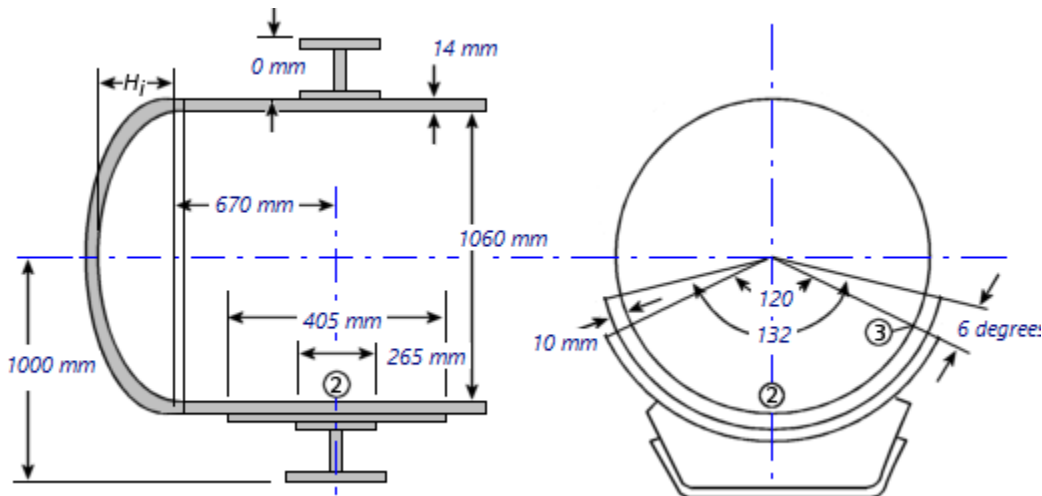
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		9018.75	Kgf
Pressure used in Analysis	P	525.195	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	11.02	136.80
Long. Stress at Bottom of Midspan	14.70	136.80
Long. Stress at Top of Saddles	20.39	136.80
Long. Stress at Bottom of Saddles	8.68	136.80
Tangential Shear in Shell	10.47	109.44
Circ. Stress at Horn of Saddle	23.24	171.00
Circ. Compressive Stress in Shell	0.83	136.80



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FileName : 192-C_BEM_Dia

Page 81 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (353/2 + 0) \cdot 1000/960$$

$$= 551.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(161, 0) \cdot 1000/2000$$

$$= 80.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(850, 0) \cdot 1000/2000$$

$$= 425.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (850/2 + 0) \cdot 1000/960$$

$$= 1329.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 7690 + \max(80.6, 552, 425, 1329)$$

$$= 9018.7 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.29 \cdot 1.2)^2)$$

$$= 1581.081 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	9197.82	Kgf
Transverse Shear Load	425.13	Kgf
Longitudinal Shear Load	850.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1060/2 + 14 + 10) \sin(120/2), 2 \cdot 538)$$

$$= 959.556 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -9019 \cdot 670 [1 - (1 - 670/3340 + (538^2 - 0^2) / (2 \cdot 670 \cdot 3340)) / (1 + (4 \cdot 0) / (3 \cdot 3340))]$$

$$= -820.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 9019 \cdot 3340 / 4 ((1 + 2(538^2 - 0^2)/(3340^2)) / (1 + (4 \cdot 0) / (3 \cdot 3340)) - 4 \cdot 670 / 3340)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 82 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

$$(3 \cdot 3340) - 4 \cdot 670 / 3340 \\ = 1879.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 525 \cdot 538 / (2 \cdot 11) - 1880 / (\pi \cdot 538^2 \cdot 11) \\ = 11.02 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 525 \cdot 538 / (2 \cdot 11) + 1880 / (\pi \cdot 538^2 \cdot 11) \\ = 14.70 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 525 \cdot 538 / (2 \cdot 11) - 821 / (0.11 \cdot \pi \cdot 538^2 \cdot 11) \\ = 20.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 525 \cdot 538 / (2 \cdot 11) + 821 / (0.19 \cdot \pi \cdot 538^2 \cdot 11) \\ = 8.68 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 9019 (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0 / 3)) \\ = 5400.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 5400 / (538 \cdot 11) \\ = 10.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{538 \cdot 11} \\ = 60.032 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(265 + 1.56 \cdot \sqrt{538 \cdot 11}, 2 \cdot 670) \\ = 385.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(137 / 137, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 9019 \cdot 0.1 / (385(11 + 1 \cdot 10)) \\ = -0.83 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\ = -9019 / (4(11 + 1 \cdot 10)385) - 12 \cdot 0.053 \cdot 9019 \cdot 538 / (3340(11 + 1 \cdot 10)^2) \\ = -23.24 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2000.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.13139 \text{E-04} \cdot 2000 (280 - 21.1)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 83 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

= 6.803 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)^3) / 12$$

$$= 5.817 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 5.82 + 57.7 (145 - 5.5)^2$$

$$= 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478368 / 171$$

$$= 145.113 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 9019$$

$$= 1835.5105 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.5916 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 533 \cdot \sin(120/2) / 1.05$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1026.4607 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 84 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

Bending Stress, $S_b = (M \cdot C_1 / I) = 2.5238 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(9019 + 179) \cdot 290) / (4 \cdot 1150 \cdot 175)}$
 $= 9.883 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10)$
 $= 959.556 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 960 / (4 - 1)$
 $= 319.852 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 320 \cdot 290 / 2$
 $= 463.786 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 9198 / (1150 \cdot 290)$
 $= 2.758 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 464 \cdot 2.76$
 $= 1279.105 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 25.5 + 16$
 $= 41.493 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1279 / 41.5$
 $= 3.023 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 265 - 10$
 $= 255.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 41.5$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 85 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

= Saddle Width / 2
= 265/2
= 132.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1551/41.5}$
= 61.135 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= $1000 - \cos(120/2) (530 + 14 + 10) - 10$
= 713.000 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$
= 122.739

Slenderness ratio [KL/r]:
= KL/r
= $1 \cdot 713/61.1$
= 11.663

Bending Moment [Rm]:
= $Fl / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= $850 / (2 \cdot 1150) \cdot 320 \cdot 713/2$
= 42.155 Kg-m.

Compressive Allowable, $KL/r < Cc$ ($11.7 < 123$) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (11.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7)^3 / (8 \cdot 123^3))$
= 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= $3.02/153 + (42.2 \cdot 132/15508023) / 175$
= 0.040

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/57.5
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 265/2
= 132.500 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(Ro + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
= $1000 - \sqrt{(554 + 10)^2 - (320/2)^2} - 10$
= 459.585 mm

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 86 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(1553/57.5)}$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 850 / (2 \cdot 1150)$$

$$= 0.370 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.37 \cdot 320 \cdot 460$$

$$= 54.344 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 54.3/117237$$

$$= 4.546 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.84 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.84)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84)^3 / (8 \cdot 123^3))$$

$$= 154.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 4.37/154 + (54.3 \cdot 132/1553) / 175$$

$$= 0.054$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9586 cm ²
Saddle Load Q0 (Weight)	Q0	7868.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	425.1 Kg
Maximum Transverse Force	Ft	425.1 Kg
Maximum Longitudinal Force	F1	850.3 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 2221.0 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2221 / (0.96 \cdot 4)$$

$$= 56.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 425 / (0.96 \cdot 4)$$

$$= 10.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 87 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 850/(82.8 • 4)
= 0.2518 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1000 • 425 + 0
= 425.14 Kg-m.

Eccentricity (e):
= Rmom / QO
= 425/7869
= 54.03 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.25, 0]
= 0.2518 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.39, Temp. Reduction = 54 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 88 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 8553.36 Kgf
Pressure used in Analysis P 525.195 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	11.11	136.80
Long. Stress at Bottom of Midspan	14.60	136.80
Long. Stress at Top of Saddles	20.00	136.80
Long. Stress at Bottom of Saddles	8.90	136.80
Tangential Shear in Shell	9.93	109.44
Circ. Stress at Horn of Saddle	22.04	171.00
Circ. Compressive Stress in Shell	0.79	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3 (353/2 + 0) \cdot 1000/960$
= 551.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
= $\max(161, 0) \cdot 1000/2000$
= 80.6 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= $\max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
= $\max(850, 0) \cdot 1000/2000$
= 425.1 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3 (850/2 + 0) \cdot 1000/960$
= 1329.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $7224 + \max(80.6, 552, 425, 1329)$
= 8553.4 Kgf

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 766 (0 + \pi/4 (1.29 \cdot 1.2)^2)$
= 1581.081 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8732.44	Kgf
Transverse Shear Load	425.13	Kgf
Longitudinal Shear Load	850.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= $\min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
= $\min(2 (1060/2 + 14 + 10) \sin(120/2), 2 \cdot 538)$
= 959.556 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 89 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / (1 + (4hm)/3L) \right]$$

$$= -8553 \cdot 670 \left[1 - \left(1 - \frac{670}{3340} + \frac{(538^2 - 0^2)}{(2 \cdot 670 \cdot 3340)} \right) / (1 + (4 \cdot 0)/(3 \cdot 3340)) \right]$$

$$= -778.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 8553 \cdot 3340 / 4 \left(\frac{(1 + 2(538^2 - 0^2)/(3340^2))}{(1 + (4 \cdot 0)/(3 \cdot 3340))} - 4 \cdot 670/3340 \right)$$

$$= 1782.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 525 \cdot 538 / (2 \cdot 11) - 1783 / (\pi \cdot 538^2 \cdot 11)$$

$$= 11.11 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 525 \cdot 538 / (2 \cdot 11) + 1783 / (\pi \cdot 538^2 \cdot 11)$$

$$= 14.60 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 525 \cdot 538 / (2 \cdot 11) - 778 / (0.11 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= 20.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 525 \cdot 538 / (2 \cdot 11) + 778 / (0.19 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= 8.90 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 8553 (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0/3))$$

$$= 5121.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 5122 / (538 \cdot 11)$$

$$= 9.93 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(538 \cdot 11)}$$

$$= 60.032 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{(538 \cdot 11)}, 2 \cdot 670)$$

$$= 385.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(137/137, 1)$$

$$= 1.0000$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 90 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 8553 \cdot 0.1 / (385(11 + 1 \cdot 10))$$

$$= -0.79 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -8553 / (4(11 + 1 \cdot 10)385) -$$

$$12 \cdot 0.053 \cdot 8553 \cdot 538 / (3340(11 + 1 \cdot 10)^2)$$

$$= -22.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)11^3) / 12$$

$$= 5.817 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.82 + 57.7(145 - 5.5)^2$$

$$= 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478368 / 171$$

$$= 145.113 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 8553$$

$$= 1740.7950 \text{ Kgf}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 91 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

Tension Stress, $St = (Fh/As) = 1.5094 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1000 - 533 \cdot \sin(120/2) / 1.05$
 $= 559.213 \text{ mm}$

Bending Moment, $M = Fh * d = 973.4937 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 2.3935 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(8553 + 179)290 / (4 * 1150 * 175))}$
 $= 9.630 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (530 + 14 + 10)$
 $= 959.556 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (N_{\text{ribs}} - 1)$
 $= 960 / (4 - 1)$
 $= 319.852 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * B_{\text{pwid}} / 2$
 $= 320 * 290 / 2$
 $= 463.786 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 8732 / (1150 * 290)$
 $= 2.618 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 464 * 2.62$
 $= 1214.386 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 25.5 + 16$
 $= 41.493 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 1214/41.5$
 $= 2.870 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 92 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9,2026

= Saddle Width - Web Thickness

= 265 - 10

= 255.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/41.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 265/2

= 132.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1551/41.5}$

= 61.135 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1000 - cos(120/2)(530 + 14 + 10) - 10

= 713.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 713/61.1

= 11.663

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 850 / (2 • 1150) • 320 • 713/2

= 42.155 Kg-m.

Compressive Allowable, KL/r < Cc (11.7 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.7)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$

= 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 2.87/153 + (42.2 • 132/15508023) / 175

= 0.039

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/57.5

= 0.000 mm

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 93 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

= Saddle Width / 2
= 265/2
= 132.500 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
= 1000 - $\sqrt{(554 + 10)^2 - (320/2)^2}$ - 10
= 459.585 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1553/57.5}$
= 51.983 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 • 460/52
= 8.841

Unit Force [Force,u]:
= FI / (2 • Baseplate Length)
= 850/(2 • 1150)
= 0.370 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force • e • L
= 0.37 • 320 • 460
= 54.344 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 54.3/117237
= 4.546 N./mm²

Compressive Allowable, KL/r < Cc (8.84 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (8.84)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84)^3 / (8 \cdot 123^3))$
= 154.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c • C1/I)/Sba
= 4.15/154 + (54.3 • 132/1553)/175
= 0.053

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9586 cm ²
Saddle Load Q0 (Weight)	Q0	7403.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	425.1 Kg
Maximum Transverse Force	Ft	425.1 Kg
Maximum Longitudinal Force	F1	850.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
= FI / (Bolt Area • Number of Bolts)



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 94 of 214

Saddle Calcs: Operating Case: Step: 20 10:11am Sep 9, 2026

$$= 850 / (0.96 \cdot 4)$$

$$= 21.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 425 / (0.96 \cdot 4)$$

$$= 10.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 850 / (82.8 \cdot 4)$$

$$= 0.2518 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1000 \cdot 425 + 0$$

$$= 425.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 425 / 7403$$

$$= 57.42 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.25, 0]$$

$$= 0.2518 \text{ cm}^2$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 10, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.39, Temp. Reduction = 54 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 192-C_BEM_Dia

Page 95 of 214

Saddle Calcs: Test Case:

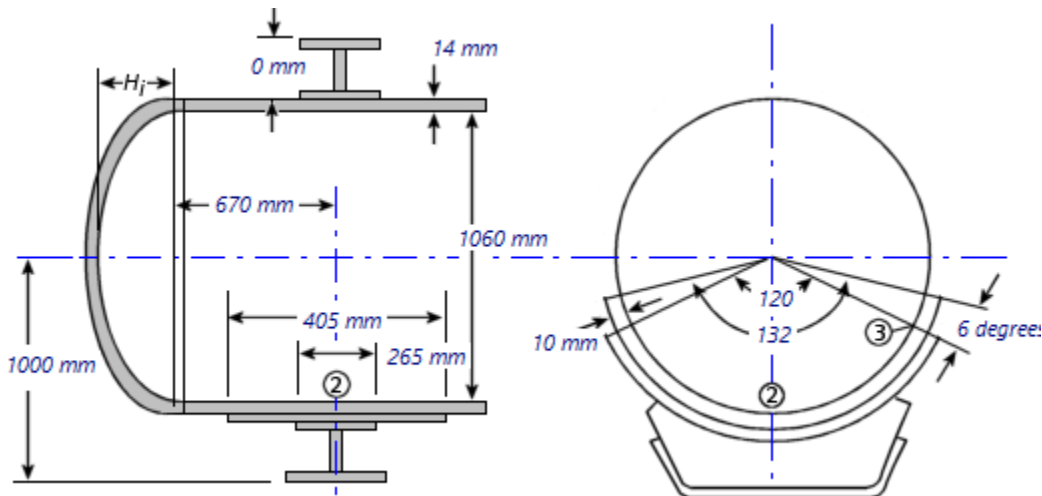
Step: 21 10:11am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	7494.77	Kgf
Pressure used in Analysis	P	1430.136 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	33.48	234.00
Long. Stress at Bottom of Midspan	36.54	234.00
Long. Stress at Top of Saddles	41.27	234.00
Long. Stress at Bottom of Saddles	31.54	234.00
Tangential Shear in Shell	8.70	138.58
Circ. Stress at Horn of Saddle	19.31	350.99
Circ. Compressive Stress in Shell	0.69	234.00



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FileName : 192-C_BEM_Dia

Page 96 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (177/2 + 0) \cdot 1000/960$$

$$= 275.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(80.6, 0) \cdot 1000/2000$$

$$= 40.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 7219 + \max(40.3, 276, 0, 0)$$

$$= 7494.8 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.29 \cdot 1.2)^2)$$

$$= 1581.081 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7673.84	Kgf
Transverse Shear Load	88.26	Kgf
Longitudinal Shear Load	80.61	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1060/2 + 14 + 10)\sin(120/2), 2 \cdot 538)$$

$$= 959.556 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -7495 \cdot 670 [1 - (1 - 670/3340 + (538^2 - 0^2)/(2 \cdot 670 \cdot 3340))/(1 + (4 \cdot 0)/(3 \cdot 3340))]$$

$$= -682.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 7495 \cdot 3340/4((1 + 2(538^2 - 0^2)/(3340^2))/(1 + (4 \cdot 0)/(3 \cdot 3340)) - 4 \cdot 670/3340)$$

$$= 1562.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 1430 \cdot 538/(2 \cdot 11) - 1562/(\pi \cdot 538^2 \cdot 11)$$

$$= 33.48 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 97 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

$$= 1430 \cdot 538 / (2 \cdot 11) + 1562 / (\pi \cdot 538^2 \cdot 11)$$

$$= 36.54 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1430 \cdot 538 / (2 \cdot 11) - 682 / (0.11 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= 41.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1430 \cdot 538 / (2 \cdot 11) + 682 / (0.19 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= 31.54 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 7495 (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0 / 3))$$

$$= 4487.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 4488 / (538 \cdot 11)$$

$$= 8.70 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(538 \cdot 11)}$$

$$= 60.032 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{(538 \cdot 11)}, 2 \cdot 670)$$

$$= 385.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 7495 \cdot 0.1 / (385(11 + 1 \cdot 10))$$

$$= -0.69 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -7495 / (4(11 + 1 \cdot 10)385) -$$

$$12 \cdot 0.053 \cdot 7495 \cdot 538 / (3340(11 + 1 \cdot 10)^2)$$

$$= -19.31 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)^3) / 12$$

$$= 5.817 \text{ cm}^4$$



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FileName : 192-C_BEM_Dia

Page 98 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9, 2026

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.82 + 57.7 (145 - 5.5)^2$$

$$= 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478368/171$$

$$= 145.113 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 7495$$

$$= 1525.3474 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.3226 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 533 \cdot \sin(120/2) / 1.05$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 853.0103 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.0973 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(7495 + 179) \cdot 290) / (4 \cdot 1150 \cdot 173)}$$

$$= 9.062 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10)$$

$$= 959.556 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 960 / (4 - 1)$$

$$= 319.852 \text{ mm}$$



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FileName : 192-C_BEM_Dia

Page 99 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 320 \cdot 290 / 2$$

$$= 463.786 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 7674 / (1150 \cdot 290)$$

$$= 2.301 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 464 \cdot 2.3$$

$$= 1067.171 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 25.5 + 16$$

$$= 41.493 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1067 / 41.5$$

$$= 2.522 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 265 - 10$$

$$= 255.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 41.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 265 / 2$$

$$= 132.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(1551 / 41.5)}$$

$$= 61.135 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1000 - \cos(120 / 2) (530 + 14 + 10) - 10$$

$$= 713.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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FileName : 192-C_BEM_Dia

Page 100 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

$$= 1 \cdot 713/61.1$$

$$= 11.663$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 80.6 / (2 \cdot 1150) \cdot 320 \cdot 713/2$$

$$= 3.997 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.52/152 + (4 \cdot 132/15508023) / 173$$

$$= 0.019$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/57.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1000 - \sqrt{ (554 + 10)^2 - (320/2)^2 } - 10$$

$$= 459.585 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ (1553/57.5) }$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 80.6 / (2 \cdot 1150)$$

$$= 0.035 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.035 \cdot 320 \cdot 460$$

$$= 5.152 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 101 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9, 2026

$$= 5.15/117237$$

$$= 0.431 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.84 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.84)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84^3) / (8 \cdot 123^3))$$

$$= 153.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 3.64/153 + (5.15 \cdot 132/1553) / 173$$

$$= 0.026$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9586 cm ²
Saddle Load QO (Weight)	QO	7397.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	40.3 Kgf
Maximum Transverse Force	Ft	88.3 Kgf
Maximum Longitudinal Force	F1	80.6 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 80.6 / (0.96 \cdot 4)$$

$$= 2.1 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 88.3 / (0.96 \cdot 4)$$

$$= 2.3 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 80.6 / (29 \cdot 4)$$

$$= 0.0682 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 88.3 + 0$$

$$= 88.26 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 88.3/7398$$

$$= 11.93 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 102 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.068, 0]

= 0.0682 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7016.37	Kgf
Pressure used in Analysis	P	1430.136	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	33.58	234.00
Long. Stress at Bottom of Midspan	36.44	234.00
Long. Stress at Top of Saddles	40.87	234.00
Long. Stress at Bottom of Saddles	31.76	234.00
Tangential Shear in Shell	8.14	138.58
Circ. Stress at Horn of Saddle	18.08	350.99
Circ. Compressive Stress in Shell	0.65	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (177/2 + 0) • 1000/960

= 275.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(80.6, 0) • 1000/2000

= 40.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 6740 + max(40.3, 276, 0, 0)



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 103 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9, 2026

= 7016.4 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (1.29 • 1.2)²)

= 1581.081 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 7195.45 Kgf

Transverse Shear Load 88.26 Kgf

Longitudinal Shear Load 80.61 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(1060/2 + 14 + 10)sin(120/2), 2•538)

= 959.556 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -7016•670 [1 - (1 - 670/3340 + (538²-0²)/

(2•670•3340))/(1+(4•0)/(3•3340))]

= -638.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)

= 7016•3340/4((1+2(538²-0²)/(3340²))/(1+(4•0)/

(3•3340)) - 4•670/3340)

= 1462.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 1430 • 538/(2•11) - 1462/(π •538²•11)

= 33.58 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 1430 • 538/(2 • 11) + 1462/(π • 538² • 11)

= 36.44 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 1430•538/(2•11)--638/(0.11• π •538²•11)

= 40.87 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 1430•538/(2•11)+-638/(0.19• π •538²•11)

= 31.76 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 7016 (3340 - 2 • 670)/(3340 + (4 • 0/3))

= 4201.4 Kgf



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 104 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 4201 / (538 \cdot 11)$$

$$= 8.14 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(538 \cdot 11)}$$

$$= 60.032 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{(538 \cdot 11)}, 2 \cdot 670)$$

$$= 385.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 7016 \cdot 0.1 / (385(11 + 1 \cdot 10))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -7016 / (4(11 + 1 \cdot 10)385) -$$

$$12 \cdot 0.053 \cdot 7016 \cdot 538 / (3340(11 + 1 \cdot 10)^2)$$

$$= -18.08 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)11^3) / 12$$

$$= 5.817 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 5.82 + 57.7(145 - 5.5)^2$$

$$= 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 105 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9, 2026

$$\begin{aligned}
 &= AY / A \\
 &= 2478368/171 \\
 &= 145.113 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2 \cdot 7016 \\
 &= 1427.9841 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 1.2382 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1000 - 533 \cdot \sin(120/2) / 1.05 \\
 &= 559.213 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 798.5623 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.9634 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(7016 + 179)290 / (4 \cdot 1150 \cdot 173)} \\
 &= 8.775 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10) \\
 &= 959.556 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 960 / (4 - 1) \\
 &= 319.852 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 320 \cdot 290/2 \\
 &= 463.786 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 7195 / (1150 \cdot 290) \\
 &= 2.158 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 464 \cdot 2.16 \\
 &= 1000.643 \text{ Kgf}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 106 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

= Rib Area + Web Area

= 25.5 + 16

= 41.493 cm²

Compressive Stress [Sc]:

= P/Ar

= 1001/41.5

= 2.365 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 265 - 10

= 255.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/41.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 265/2

= 132.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1551/41.5}$

= 61.135 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1000 - cos(120/2)(530 + 14 + 10) - 10

= 713.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 713/61.1

= 11.663

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 80.6 / (2 • 1150) • 320 • 713/2

= 3.997 Kg-m.

Compressive Allowable, KL/r < Cc (11.7 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$

= 152.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 107 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am Sep 9,2026

$$= 2.37/152 + (4 \cdot 132/15508023)/173$$

$$= 0.017$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/57.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1000 - \sqrt{(554 + 10)^2 - (320/2)^2} - 10$$

$$= 459.585 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1553/57.5)}$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 80.6 / (2 \cdot 1150)$$

$$= 0.035 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.035 \cdot 320 \cdot 460$$

$$= 5.152 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5.15/117237$$

$$= 0.431 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.84 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.84)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84)^3 / (8 \cdot 123^3))$$

$$= 153.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 3.42/153 + (5.15 \cdot 132/1553)/173$$

$$= 0.025$$

Input Data for Base Plate Bolting Calculations:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 108 of 214

Saddle Calcs: Test Case:

Step: 21 10:11am

Sep 9, 2026

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9586 cm ²
Saddle Load Q0 (Weight)	Q0	6919.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	40.3 Kg
Maximum Transverse Force	Ft	88.3 Kg
Maximum Longitudinal Force	Fl	80.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 80.6 / (0.96 \cdot 4)$$

$$= 2.1 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 88.3 / (0.96 \cdot 4)$$

$$= 2.3 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 80.6 / (29 \cdot 4)$$

$$= 0.0682 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 88.3 + 0$$

$$= 88.26 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 88.3 / 6920$$

$$= 12.75 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.068, 0]$$

$$= 0.0682 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 109 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

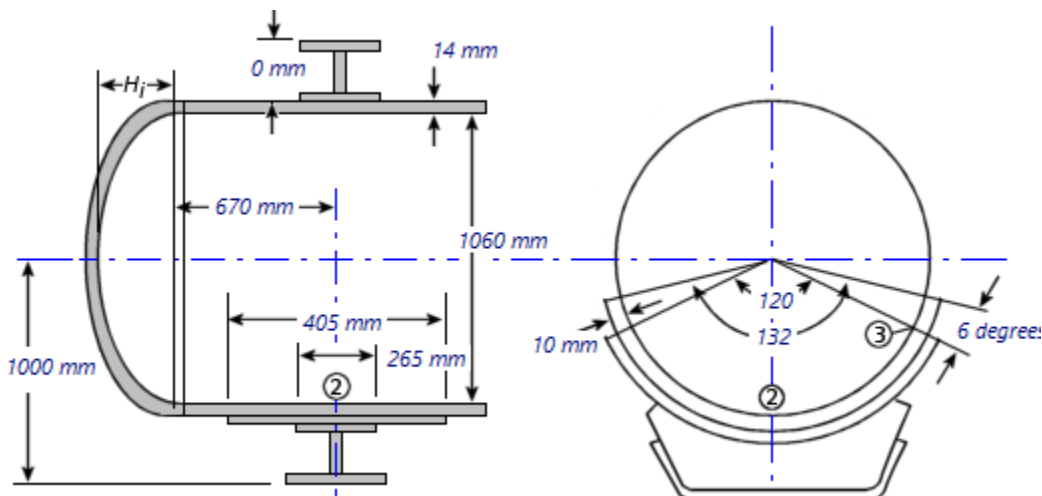
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	9018.75	Kgf
Pressure used in Analysis	P	-101.130 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-4.31	-187.92	
Long. Stress at Top of Midspan	4.31	136.80	
Long. Stress at Bottom of Midspan	-0.64	-187.92	
Long. Stress at Bottom of Midspan	0.64	136.80	
Long. Stress at Top of Saddles	5.06	136.80	
Long. Stress at Bottom of Saddles	-6.65	-187.92	
Long. Stress at Bottom of Saddles	6.65	136.80	



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 110 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9, 2026

Tangential Shear in Shell	10.47	109.44
Circ. Stress at Horn of Saddle	23.24	171.00
Circ. Compressive Stress in Shell	0.83	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (353/2 + 0) \cdot 1000/960$$

$$= 551.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(161, 0) \cdot 1000/2000$$

$$= 80.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(850, 0) \cdot 1000/2000$$

$$= 425.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (850/2 + 0) \cdot 1000/960$$

$$= 1329.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 7690 + \max(80.6, 552, 425, 1329)$$

$$= 9018.7 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.29 \cdot 1.2)^2)$$

$$= 1581.081 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	9197.82	Kgf
Transverse Shear Load	425.13	Kgf
Longitudinal Shear Load	850.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1060/2 + 14 + 10) \sin(120/2), 2 \cdot 538)$$

$$= 959.556 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -9019 \cdot 670 [1 - (1 - 670/3340 + (538^2 - 0^2) /$$

$$(2 \cdot 670 \cdot 3340) / (1 + (4 \cdot 0) / (3 \cdot 3340))]$$

$$= -820.6 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 111 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9,2026

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$$

$$= 9019 \cdot 3340 / 4 \cdot ((1 + 2(538^2 - 0^2) / (3340^2)) / (1 + (4 \cdot 0) / (3 \cdot 3340)) - 4 \cdot 670 / 3340)$$

$$= 1879.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -101 \cdot 538 / (2 \cdot 11) - 1880 / (\pi \cdot 538^2 \cdot 11)$$

$$= -4.31 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 538 / (2 \cdot 11) + 1880 / (\pi \cdot 538^2 \cdot 11)$$

$$= -0.64 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -101 \cdot 538 / (2 \cdot 11) - 821 / (0.11 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= 5.06 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 538 / (2 \cdot 11) + 821 / (0.19 \cdot \pi \cdot 538^2 \cdot 11)$$

$$= -6.65 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 9019 \cdot (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0 / 3))$$

$$= 5400.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 5400 / (538 \cdot 11)$$

$$= 10.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{538 \cdot 11}$$

$$= 60.032 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(265 + 1.56 \cdot \sqrt{538 \cdot 11}, 2 \cdot 670)$$

$$= 385.06 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(137/137, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 9019 \cdot 0.1 / (385(11 + 1 \cdot 10))$$

$$= -0.83 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -9019 / (4(11 + 1 \cdot 10)385) - 12 \cdot 0.053 \cdot 9019 \cdot 538 / (3340(11 + 1 \cdot 10)^2)$$

$$= -23.24 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 112 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)11^3)/12$$

$$= 5.817 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.82 + 57.7(145 - 5.5)^2$$

$$= 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478368/171$$

$$= 145.113 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 9019$$

$$= 1835.5105 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.5916 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 533 \cdot \sin(120/2) / 1.05$$

$$= 559.213 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1026.4607 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5238 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 113 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9, 2026

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(9019 + 179) \cdot 290}{4 \cdot 1150 \cdot 175}}$$

$$= 9.883 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10)$$

$$= 959.556 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 960 / (4 - 1)$$

$$= 319.852 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 320 \cdot 290 / 2$$

$$= 463.786 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 9198 / (1150 \cdot 290)$$

$$= 2.758 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 464 \cdot 2.76$$

$$= 1279.105 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.5 + 16$$

$$= 41.493 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1279 / 41.5$$

$$= 3.023 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 265 - 10$$

$$= 255.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 41.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265 / 2$$

$$= 132.500 \text{ mm}$$

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 114 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1551/41.5}$$

$$= 61.135 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (530 + 14 + 10) - 10$$

$$= 713.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943008/262}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 713/61.1$$

$$= 11.663$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 850 / (2 \cdot 1150) \cdot 320 \cdot 713/2$$

$$= 42.155 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7)^3 / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.02/153 + (42.2 \cdot 132/15508023) / 175$$

$$= 0.040$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/57.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1000 - \sqrt{(554 + 10)^2 - (320/2)^2} - 10$$

$$= 459.585 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1553/57.5}$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 115 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 850 / (2 \cdot 1150)$$

$$= 0.370 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.37 \cdot 320 \cdot 460$$

$$= 54.344 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 54.3/117237$$

$$= 4.546 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.84 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.84)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84^3) / (8 \cdot 123^3))$$

$$= 154.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 4.37/154 + (54.3 \cdot 132/1553) / 175$$

$$= 0.054$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	538.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	3340.00	mm
Distance from Saddle to Vessel tangent	a1 or a	670.00	mm
Saddle Width	b1 or b	265.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	405.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	136.80	N./mm ²
Shell Allowable Stress used in Calculation		136.80	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		8553.36	Kgf
Pressure used in Analysis	P	-101.130	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-4.22	-187.92
Long. Stress at Top	of Midspan	4.22	136.80



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 116 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9, 2026

Long. Stress at Bottom of Midspan	-0.73	-187.92
Long. Stress at Bottom of Midspan	0.73	136.80
Long. Stress at Top of Saddles	4.67	136.80
Long. Stress at Bottom of Saddles	-6.44	-187.92
Long. Stress at Bottom of Saddles	6.44	136.80
Tangential Shear in Shell	9.93	109.44
Circ. Stress at Horn of Saddle	22.04	171.00
Circ. Compressive Stress in Shell	0.79	136.80

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (353/2 + 0) \cdot 1000/960$$

$$= 551.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(161, 0) \cdot 1000/2000$$

$$= 80.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(850, 0) \cdot 1000/2000$$

$$= 425.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (850/2 + 0) \cdot 1000/960$$

$$= 1329.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 7224 + \max(80.6, 552, 425, 1329)$$

$$= 8553.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.29 \cdot 1.2)^2)$$

$$= 1581.081 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8732.44	Kgf
Transverse Shear Load	425.13	Kgf
Longitudinal Shear Load	850.27	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1060/2 + 14 + 10)\sin(120/2), 2 \cdot 538)$$

$$= 959.556 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 117 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9, 2026

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right] \\
 &= -8553 \cdot 670 \left[1 - \left(1 - \frac{670}{3340} + \frac{(538^2 - 0^2)}{(2 \cdot 670 \cdot 3340)} \right) / \left(1 + \frac{(4 \cdot 0)}{3 \cdot 3340} \right) \right] \\
 &= -778.3 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 \left(\left(1 + 2 \frac{(Rm^2 - hm^2)}{(L^2)} \right) / \left(1 + \frac{(4hm^2)}{(3L)} \right) - 4a/L \right) \\
 &= 8553 \cdot 3340 / 4 \left(\left(1 + 2 \frac{(538^2 - 0^2)}{(3340^2)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 3340)} \right) - 4 \cdot 670 / 3340 \right) \\
 &= 1782.6 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\
 &= -101 \cdot 538 / (2 \cdot 11) - 1783 / (\pi \cdot 538^2 \cdot 11) \\
 &= -4.22 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -101 \cdot 538 / (2 \cdot 11) + 1783 / (\pi \cdot 538^2 \cdot 11) \\
 &= -0.73 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\
 &= -101 \cdot 538 / (2 \cdot 11) - 778 / (0.11 \cdot \pi \cdot 538^2 \cdot 11) \\
 &= 4.67 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -101 \cdot 538 / (2 \cdot 11) + 778 / (0.19 \cdot \pi \cdot 538^2 \cdot 11) \\
 &= -6.44 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 8553 (3340 - 2 \cdot 670) / (3340 + (4 \cdot 0 / 3)) \\
 &= 5121.8 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 5122 / (538 \cdot 11) \\
 &= 9.93 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(538 \cdot 11)} \\
 &= 60.032 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(265 + 1.56 \cdot \sqrt{(538 \cdot 11)}, 2 \cdot 670) \\
 &= 385.06 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(137/137, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 8553 \cdot 0.1 / (385(11 + 1 \cdot 10)) \\
 &= -0.79 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -8553 / (4(11 + 1 \cdot 10)385) -
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 118 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

$$12 \cdot 0.053 \cdot 8553 \cdot 538 / (3340(11 + 1 \cdot 10)^2) \\ = -22.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1150.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	290.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	470.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\ = ((1.56 \cdot \sqrt{(533 \cdot 11)} + 405)^3) / 12 \\ = 5.817 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 5.82 + 57.7(145 - 5.5)^2 \\ = 11250.438 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	524.4	11.0	5.5	57.7	31729.2	0.113E+05
Wearplate	405.0	10.0	16.0	40.5	64800.0	0.675E+04
Web	10.0	436.0	239.0	43.6	1042039.9	0.108E+05
BasePlate	290.0	10.0	462.0	29.0	1339799.8	0.291E+05
Totals	...	...	...	170.8	2478368.8	0.579E+05

Distance to Centroid [C1]:

$$= AY / A \\ = 2478368 / 171 \\ = 145.113 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2 \\ = 180 - 120 / 2 \\ = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ = (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.2 \cdot 8553 \\ = 1740.7950 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.5094 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ = 1000 - 533 \cdot \sin(120/2) / 1.05 \\ = 559.213 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 119 of 214

Saddle Calcs: Ext. Pressure: Step: 22 10:11am Sep 9,2026

Bending Moment, $M = F_h \cdot d = 973.4937 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 2.3935 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(8553 + 179) \cdot 290) / (4 \cdot 1150 \cdot 175)}$

$= 9.630 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (530 + 14 + 10)$

$= 959.556 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 960 / (4 - 1)$

$= 319.852 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 320 \cdot 290 / 2$

$= 463.786 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 8732 / (1150 \cdot 290)$

$= 2.618 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 464 \cdot 2.62$

$= 1214.386 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 25.5 + 16$

$= 41.493 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 1214 / 41.5$

$= 2.870 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	265.0	...	26.5	...	0.155E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 265 - 10$

$= 255.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 41.5$

$= 0.000 \text{ mm}$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 120 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1551/41.5}$$

$$= 61.135 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (530 + 14 + 10) - 10$$

$$= 713.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199943008/262}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 713/61.1$$

$$= 11.663$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 850 / (2 \cdot 1150) \cdot 320 \cdot 713/2$$

$$= 42.155 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.87/153 + (42.2 \cdot 132/15508023) / 175$$

$$= 0.039$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	255.0	0.0	25.5	0.0	0.155E+04
Web	319.9	10.0	0.0	32.0	0.0	2.67
Totals	...	...	...	57.5	...	0.155E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/57.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 265/2$$

$$= 132.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1000 - \sqrt{(554 + 10)^2 - (320/2)^2} - 10$$

$$= 459.585 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 121 of 214

Saddle Calcs: Ext. Pressure:

Step: 22 10:11am

Sep 9, 2026

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1553/57.5)}$$

$$= 51.983 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 460/52$$

$$= 8.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 850 / (2 \cdot 1150)$$

$$= 0.370 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.37 \cdot 320 \cdot 460$$

$$= 54.344 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 54.3/117237$$

$$= 4.546 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.84 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.84)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (8.84) / (8 \cdot 123) - (8.84^3) / (8 \cdot 123^3))$$

$$= 154.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 4.15/154 + (54.3 \cdot 132/1553)/175$$

$$= 0.053$$

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 192-C_BEM_Dia

Page 122 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am Sep 9,2026

Input, Nozzle Desc: Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	900.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	433.0500	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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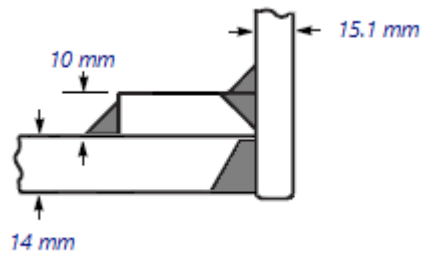
FileName : 192-C_BEM_Dia

Page 123 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026



Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.562 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 900 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 900)$
 $= 3.5156 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 900 \cdot 124 / (118 \cdot 1 - 0.6 \cdot 900)$
 $= 0.9535 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	497.7496 mm
Parallel to Vessel Wall, opening length	d	248.8748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.861$



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FileName : 192-C_BEM_Dia

Page 124 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

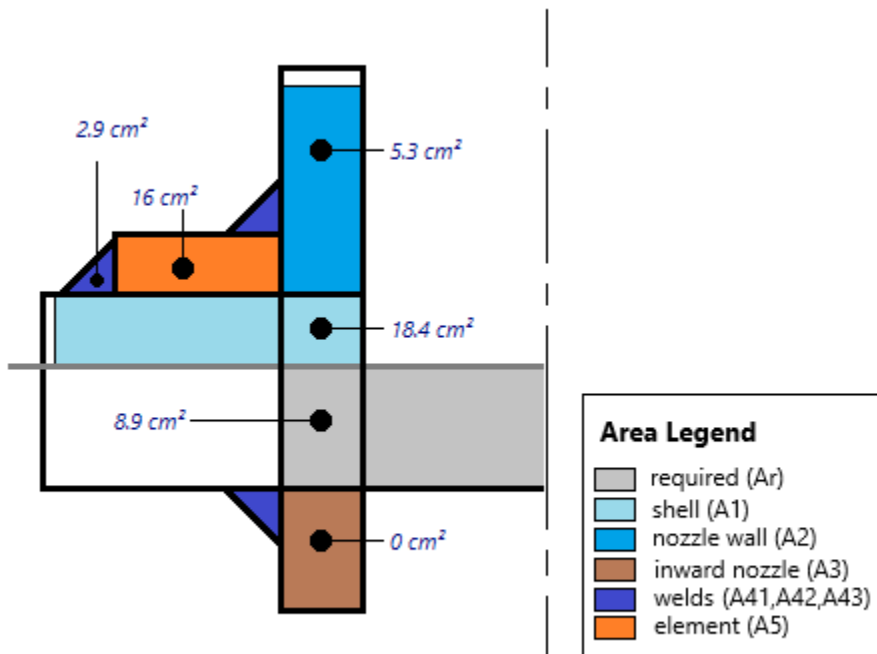


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.867	NA	NA
Area in Shell	A1	18.376	NA	NA
Area in Nozzle Wall	A2	5.274	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.938	NA	NA
Area in Element	A5	16.000	NA	NA
TOTAL AREA AVAILABLE	Atot	42.588	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (249 \cdot 3.52 \cdot 1 + 2 \cdot 12.1 \cdot 3.52 \cdot 1 \cdot (1 - 0.86)) \\
 &= 8.867 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 249(1 \cdot 11 - 1 \cdot 3.52) - 2 \cdot 12.1 \\
 &\quad (1 \cdot 11 - 1 \cdot 3.52) \cdot (1 - 0.86) \\
 &= 18.376 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (12.1 - 0.95) \cdot 0.86
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 125 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

$$= 5.274 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 15^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 10^2 \cdot 1$$

$$= 2.938 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (433 - 273) \cdot 10 \cdot 1$$

$$= 16.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.9535 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 6.5156 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 6.5156 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.1, \max(6.52, 5.5)]$$

$$= 6.5156 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.95, 6.52)$$

$$= 6.5156 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.5,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	272.5 N./mm ²	Passed
Occasional	:	4.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	14.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.2$, $tr = 0.95$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.093$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $0 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.32$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10$, $c = 3 \text{ mm}$, $E^* = 1$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 126 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.32$, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 10$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.32$, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 13.2$, $tr = 0.95$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.093$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $900.00 / 1960.01 = 0.459$

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 \cdot Wo \text{ mm}$
Pad Weld	$5.0000 = 0.5 \cdot TminPad$	$7.0700 = 0.7 \cdot Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (8.87 - 18.4 + 2 \cdot 12.1 \cdot 0.86 \cdot (1 \cdot 11 - 3.52)) \cdot 137)$
 $= \max(0, -11106.31) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (5.27 + 16 + 2.94 - 0 \cdot 0.86) \cdot 137$
 $= 33823.94 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (5.27 + 0 + 1.94 + (2.29)) \cdot 137$
 $= 13275.15 \text{ Kg}$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 127 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am Sep 9, 2026

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (5.27 + 0 + 2.94 + 16 + (2.29)) \cdot 137$$

$$= 37023.63 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 273 \cdot 15 \cdot 0.49 \cdot 118$$

$$= 37931. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 433 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 46563. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 130) \cdot (15.1 - 3) \cdot 0.7 \cdot 118$$

$$= 41733. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 273 \cdot 10 \cdot 0.74 \cdot 137$$

$$= 44338. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 273 \cdot (14 - 3) \cdot 0.74 \cdot 137$$

$$= 48772. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (46563 + 41733) = 88296 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (37931 + 44338 + 48772 + 0) = 131042 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (46563 + 48772 + 0) = 95335 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 88296	Kgf, must exceed W = 0 Kgf	or W1 = 33823 Kgf
Path 2-2 = 131041	Kgf, must exceed W = 0 Kgf	or W2 = 13275 Kgf
Path 3-3 = 95335	Kgf, must exceed W = 0 Kgf	or W3 = 37023 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2572 KPa.

The Drop for this Nozzle is : 17.8858 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 281.8858 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 248.87 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 128 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	242.875	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	433.050	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	900.00	KPa.
Radial Load	P	-1910.99	Kgf
Circumferential Shear	Vc	1274.00	Kgf
Longitudinal Shear	Vl	1274.00	Kgf
Circumferential Moment	Mc	892.00	Kg-m.
Longitudinal Moment	Ml	892.00	Kg-m.
Torsional Moment	Mt	1274.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)} \\
 &= 273.05 + 2 \cdot 1.65 \cdot \sqrt{543.5(14.0 - 3.0)} \\
 &= 528.209 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1087.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	21.000	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	1.737	
d / t	=	22.589	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.807$$

Nr/P	=	0.071
Mr/P	=	0.096



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 129 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

MØ/P = 0.047
 NØ/P = 0.141
 MrD/Mc = 0.212
 NrDL/Mc = 0.119
 MØd/Mc = 0.123
 NØDL/Mc = 0.142
 MrD/Ml = 0.129
 NrDL/Ml = 0.074
 MØD/Ml = 0.072
 NØDL/Ml = 0.241

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	22	23	22	23
Outplane Stress Summary	-8	31	89	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	-22	40	100	-61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	25	43	101	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 130 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am Sep 9,2026

Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-53	53	53	-53
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11
Outplane Stress Summary	-34	48	93	-38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-92	92	92	-92
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22	23	22	23
Inplane Stress Summary	-51	85	151	-81

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	52	86	151	82

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11
Hoop Stress Summary	0	0	34	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	46	-46	46	-46
Axial Membrane (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM Dia

Page 131 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am Sep 9, 2026

Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-120	120	120	-120
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-81	67	188	-145
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8

Shear Stress Summary	11	11	6	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	83	70	188	181
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11

Hoop Stress Summary	6	7	26	28
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	46	-46	46	-46
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-200	200	200	-200
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-161	147	268	-226
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary	6	6	11	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 132 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	168	147	269	255

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{538.5(14.0 - 3.0)}$$

$$= 527.032 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1077.000 mm
Nozzle Outside Diameter	(d) =	433.050 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	92.088 mm
T / t	=	0.119
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.979$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.084
M0/P	=	0.024
N0/P	=	0.038
MrD/Mc	=	0.253
NrDL/Mc	=	0.101
M0d/Mc	=	0.076
N0DL/Mc	=	0.065
MrD/Ml	=	0.063
NrDL/Ml	=	0.058
M0D/Ml	=	0.019
N0DL/Ml	=	0.042

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	43	44	43	44
Outplane Stress Summary	45	39	97	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	77	-77	77	-77
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	21	21	21	21



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 133 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am Sep 9,2026

Inplane Stress Summary	34	4	179	-101
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	48	40	179	116
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-75	75	75	-75
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-36	69	136	-60
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	77	-77	77	-77
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-253	253	253	-253
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	44	43	44

Inplane Stress Summary	-141	211	399	-262
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left	Left	Right	Right



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 134 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

	Outside	Inside	Outside	Inside
Two * Max Shear Stress	141	211	399	262

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	-12	-12	22	22	-5	-5	11	11
Circ. Q (SUS)	-20	20	42	-42	-68	68	116	-116
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-2	-2	8	8	-5	-5	15	15
Long. Q (SUS)	-32	32	80	-80	-42	42	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	10.6	11.4	44.0	45.0	17.1	18.1	33.1	34.1
Pm+Pl+Q (Total)	25.3	43.8	100.1	64.5	52.0	86.4	149.6	82.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	137.00	Passed
Pm+Pl (SUS)	45.04	205.50	Passed
Pm+Pl+Q (TOTAL)	149.60	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-1	-1	11	11	-9	-9	23	23
Circ. Q (SUS)	4	-4	40	-40	-176	176	330	-330
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	16	16	-5	-5	15	15
Long. Q (SUS)	15	-15	139	-139	-53	53	97	-97
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+Pl (SUS)	42.0	43.0	54.1	55.1	34.1	35.1	66.0	67.0



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 135 of 214

Nozzle Calcs.: Tube inlet

Nozl: 13 10:11am

Sep 9, 2026

Pm+Pl+Q (Total)	47.2	39.5	176.0	117.1	142.0	211.0	396.1	263.1
-----------------	------	------	-------	-------	-------	-------	-------	-------

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	137.00	Passed
Pm+Pl (SUS)	67.03	205.50	Passed
Pm+Pl+Q (TOTAL)	396.06	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-12	-12	22	22	-5	-5	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-74	74	166	-166	-154	154	246	-246
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	8.6	8.5	32.3	33.3	13.6	14.6	25.7	26.6
Pm+Pl+Q (Total)	83.2	69.9	186.2	179.4	167.4	146.3	266.4	252.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	33.30	177.00	Passed
Pm+Pl+Q (TOTAL)	266.41	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 136 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

Input, Nozzle Desc: Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	910.390	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	433.0500	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



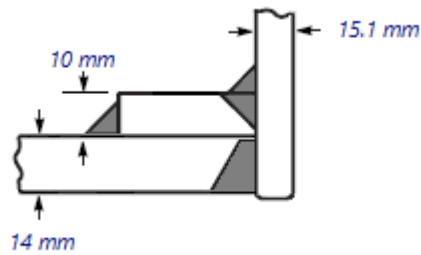
PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 137 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9, 2026



Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.562 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 910 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 910)$
 $= 3.5563 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 910 \cdot 124 / (118 \cdot 1 - 0.6 \cdot 910)$
 $= 0.9646 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	497.7496 mm
Parallel to Vessel Wall, opening length	d	248.8748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.861$

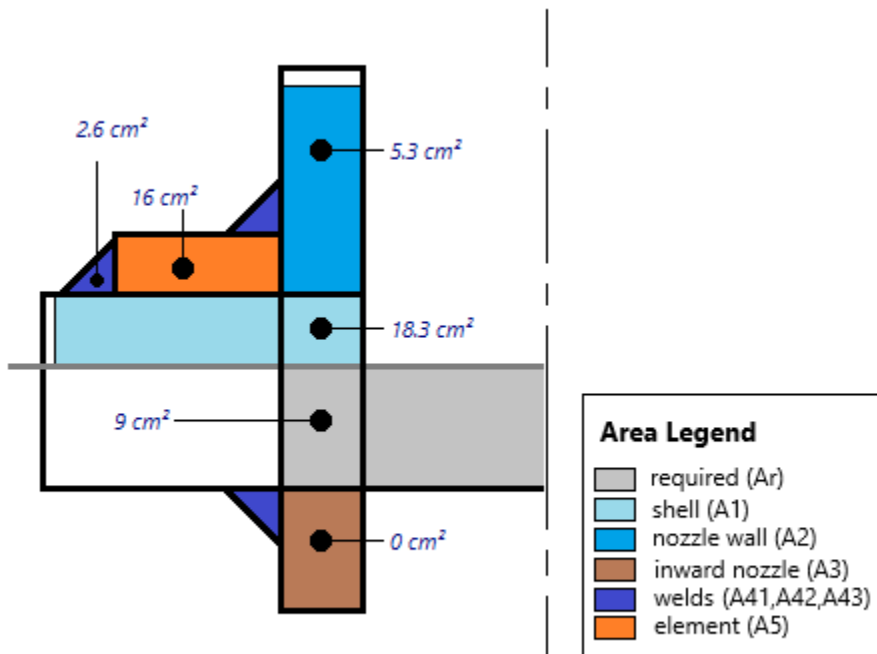


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.970	NA	NA
Area in Shell	A1	18.276	NA	NA
Area in Nozzle Wall	A2	5.269	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.578	NA	NA
Area in Element	A5	16.000	NA	NA
TOTAL AREA AVAILABLE	Atot	42.123	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (249 \cdot 3.56 \cdot 1 + 2 \cdot 12.1 \cdot 3.56 \cdot 1 \cdot (1 - 0.86)) \\
 &= 8.970 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 249(1 \cdot 11 - 1 \cdot 3.56) - 2 \cdot 12.1 \\
 &\quad (1 \cdot 11 - 1 \cdot 3.56) \cdot (1 - 0.86) \\
 &= 18.276 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 27.5) \cdot (12.1 - 0.96) \cdot 0.86
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 139 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am

Sep 9, 2026

= 5.269 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $15^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 8^2 \cdot 1$

= 2.578 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$

= $(433 - 273) \cdot 10 \cdot 1$

= 16.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.9646$ mm

Wall Thickness per UG16(b), $tr_{16b} = 5.5000$ mm

Wall Thickness, shell/head, internal pressure $tr_{b1} = 6.5563$ mm

Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 6.5563$ mm

Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000$ mm

Wall Thickness per table UG-45 $tb_3 = 11.1026$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[11.1, \max(6.56, 5.5)]$

= 6.5563 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(3.96, 6.56)$

= 6.5563 mm

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	272.4 N./mm ²	Passed
Occasional	:	4.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	14.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 13.2$, $tr = 0.96$, $c = 3$ mm, $E* = 1$

Thickness Ratio = $tr \cdot E* / (t_g - c) = 0.095$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E* = 1$

Thickness Ratio = $tr \cdot E* / (t_g - c) = 0.32$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 140 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am

Sep 9, 2026

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.32$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.32$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 13.2$, $tr = 0.96$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.095$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $910.39 / 1580.01 = 0.576$

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 10.0000 mm

Intermediate Calc. for pad/shell Welds T_{minPad} 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 \cdot W_o$ mm
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (8.97 - 18.3 + 2 \cdot 12.1 \cdot 0.86 \cdot (1 \cdot 11 - 3.56)) \cdot 137)$$

$$= \max(0, -10834.83) \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_r2) \cdot S_v$$

$$= (5.27 + 16 + 2.58 - 0 \cdot 0.86) \cdot 137$$

$$= 33313.71 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (5.27 + 0 + 1.94 + (2.29)) \cdot 137$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 141 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9, 2026

= 13267.83 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2*tn*t*fr1))*S

= (5.27 + 0 + 2.58 + 16 + (2.29)) * 137

= 36513.40 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) * Dlo * Wo * 0.49 * Snw

= (3.14/2.0) * 273 * 15 * 0.49 * 118

= 37931. Kgf

Shear, Pad Element Weld [Spew]:

= (π/2) * DP * WP * 0.49 * SEW

= (3.14/2.0) * 433 * 8 * 0.49 * 137

= 37250. Kgf

Shear, Nozzle Wall [Snw]:

= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn

= (3.14 * 130) * (15.1 - 3) * 0.7 * 118

= 41733. Kgf

Tension, Pad Groove Weld [Tpgw]:

= (π/2) * Dlo * Wgpn * 0.74 * Seg

= (3.14/2) * 273 * 10 * 0.74 * 137

= 44338. Kgf

Tension, Shell Groove Weld [Tngw]:

= (π/2) * Dlo * Wgnvi * 0.74 * Sng

= (3.14/2.0) * 273 * (8 - 0) * 0.74 * 137

= 35471. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (37250 + 41733) = 78984 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (37931 + 44338 + 35471 + 0) = 117740 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (37250 + 35471 + 0) = 72721 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 78983 Kgf, must exceed W = 0 Kgf or W1 = 33313 Kgf

Path 2-2 = 117740 Kgf, must exceed W = 0 Kgf or W2 = 13267 Kgf

Path 3-3 = 72721 Kgf, must exceed W = 0 Kgf or W3 = 36513 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2543 KPa.

The Drop for this Nozzle is : 17.8858 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 281.8858 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell

ID

Shell Internal Corrosion Allowance

3.0000 mm

Shell External Corrosion Allowance

0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 142 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	242.875	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	433.050	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	910.39	KPa.
Radial Load	P	-1910.99	Kgf
Circumferential Shear	Vc	1274.00	Kgf
Longitudinal Shear	Vl	1274.00	Kgf
Circumferential Moment	Mc	892.00	Kg-m.
Longitudinal Moment	Ml	892.00	Kg-m.
Torsional Moment	Mt	1274.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{543.5(14.0 - 3.0)}$
 $= 528.209 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1087.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	21.000	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	1.737	
d / t	=	22.589	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.807$$

Nr/P	=	0.071
Mr/P	=	0.096
M0/P	=	0.047
N0/P	=	0.141
MrD/Mc	=	0.212



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 143 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

NrDL/Mc = 0.119
MØd/Mc = 0.123
NØDL/Mc = 0.142
MrD/Ml = 0.129
NrDL/Ml = 0.074
MØD/Ml = 0.072
NØDL/Ml = 0.241

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	22	23	22	23
Outplane Stress Summary	-8	31	89	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	-22	40	100	-60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	25	44	101	65

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-53	53	53	-53



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 144 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11

Outplane Stress Summary	-34	48	93	-37
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-92	92	92	-92
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22	23	22	23

Inplane Stress Summary	-50	85	151	-81
------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	51	86	151	82
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11

Hoop Stress Summary	0	0	34	35
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	46	-46	46	-46
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-120	120	120	-120



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FileName : 192-C_BEM_Dia

Page 145 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am

Sep 9, 2026

Axial Pressure Stress	5	5	5	5
-----------------------	---	---	---	---

Axial Stress Summary	-81	67	188	-145
----------------------	-----	----	-----	------

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
-------------------	---	---	----	----

Shear due to (Vl)	0	0	0	0
-------------------	---	---	---	---

Shear due to Torsion	8	8	8	8
----------------------	---	---	---	---

Shear Stress Summary	11	11	6	6
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	83	70	188	181
------------------------	----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
--------------------	---	---	---	---

Hoop Bending (P)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (Mc)	-10	-10	10	10
--------------------	-----	-----	----	----

Hoop Bending (Mc)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (ML)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (ML)	0	0	0	0
-------------------	---	---	---	---

Hoop Pressure Stress	10	11	10	11
----------------------	----	----	----	----

Hoop Stress Summary	6	7	27	28
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
---------------------	---	---	---	---

Axial Bending (P)	46	-46	46	-46
--------------------	----	-----	----	-----

Axial Membrane (Mc)	-14	-14	14	14
---------------------	-----	-----	----	----

Axial Bending (Mc)	-200	200	200	-200
--------------------	------	-----	-----	------

Axial Membrane (ML)	0	0	0	0
---------------------	---	---	---	---

Axial Bending (ML)	0	0	0	0
--------------------	---	---	---	---

Axial Pressure Stress	5	5	5	5
-----------------------	---	---	---	---

Axial Stress Summary	-161	147	268	-225
----------------------	------	-----	-----	------

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---

Shear due to (Vl)	-2	-2	2	2
-------------------	----	----	---	---

Shear due to Torsion	8	8	8	8
----------------------	---	---	---	---

Shear Stress Summary	6	6	11	11
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	168	147	269	255
------------------------	-----	-----	-----	-----



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 146 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(538.5(14.0 - 3.0))}$$

$$= 527.032 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1077.000 mm
 Nozzle Outside Diameter (d) = 433.050 mm
 Vessel Thickness used (T) = 11.000 mm
 Nozzle Thickness used (t) = 92.088 mm
 T / t = 0.119
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.979$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.051
 Mr/P = 0.084
 M0/P = 0.024
 N0/P = 0.038
 MrD/Mc = 0.253
 NrDL/Mc = 0.101
 M0d/Mc = 0.076
 N0DL/Mc = 0.065
 MrD/Ml = 0.063
 NrDL/Ml = 0.058
 M0D/Ml = 0.019
 N0DL/Ml = 0.042

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	43	44	43	44
Outplane Stress Summary	46	39	98	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	77	-77	77	-77
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	35	5	180	-101



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 147 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am Sep 9,2026

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			48	40	180	117

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			22	-22	22	-22
Outplane Membrane (Mc)			-10	-10	10	10
Outplane Bending (Mc)			-75	75	75	-75
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			21	21	21	21
Outplane Stress Summary			-36	70	137	-59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			7	7	7	7
Inplane Bending (P)			77	-77	77	-77
Inplane Membrane (Mc)			-16	-16	16	16
Inplane Bending (Mc)			-253	253	253	-253
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			43	44	43	44
Inplane Stress Summary			-141	211	400	-262

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			3	3	3	3
Shear Stress Summary			2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			141	211	400	262



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 148 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am

Sep 9,2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	-12	-12	22	22	-5	-5	11	11
Circ. Q (SUS)	-20	20	42	-42	-68	68	116	-116
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-2	-2	8	8	-5	-5	15	15
Long. Q (SUS)	-32	32	80	-80	-42	42	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	10.6	11.4	44.0	45.0	17.1	18.1	33.1	34.1
Pm+Pl+Q (Total)	25.3	43.8	100.1	64.5	52.0	86.4	149.6	82.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	137.00	Passed
Pm+Pl (SUS)	45.04	205.50	Passed
Pm+Pl+Q (TOTAL)	149.60	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-1	-1	11	11	-9	-9	23	23
Circ. Q (SUS)	4	-4	40	-40	-176	176	330	-330
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	16	16	-5	-5	15	15
Long. Q (SUS)	15	-15	139	-139	-53	53	97	-97
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+Pl (SUS)	42.0	43.0	54.1	55.1	34.1	35.1	66.0	67.0
Pm+Pl+Q (Total)	47.2	39.5	176.0	117.1	142.0	211.0	396.1	263.1



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FileName : 192-C_BEM_Dia

Page 149 of 214

Nozzle Calcs.: Tube outlet

Nozl: 14 10:11am

Sep 9, 2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	137.00	Passed
Pm+Pl (SUS)	67.03	205.50	Passed
Pm+Pl+Q (TOTAL)	396.06	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-12	-12	22	22	-5	-5	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-74	74	166	-166	-154	154	246	-246
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	8.6	8.5	32.3	33.3	13.6	14.6	25.7	26.6
Pm+Pl+Q (Total)	83.2	69.9	186.2	179.4	167.4	146.3	266.4	252.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	33.30	177.00	Passed
Pm+Pl+Q (TOTAL)	266.41	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 192-C_BEM_Dia

Page 150 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am Sep 9,2026

Input, Nozzle Desc: Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	530.390	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	3340.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3931.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	136.80	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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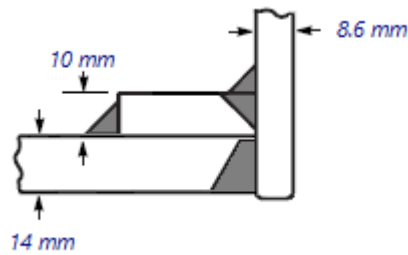
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Page 151 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026



Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.826 in.
Actual Thickness Used in Calculation	0.337 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 530 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 530)$
 $= 2.0715 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 530 \cdot 51.6 / (118 \cdot 1 - 0.6 \cdot 530)$
 $= 0.2325 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5089 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	206.3608 mm
Parallel to Vessel Wall, opening length	d	103.1804 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		23.8995 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.863$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.863$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.863$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 152 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

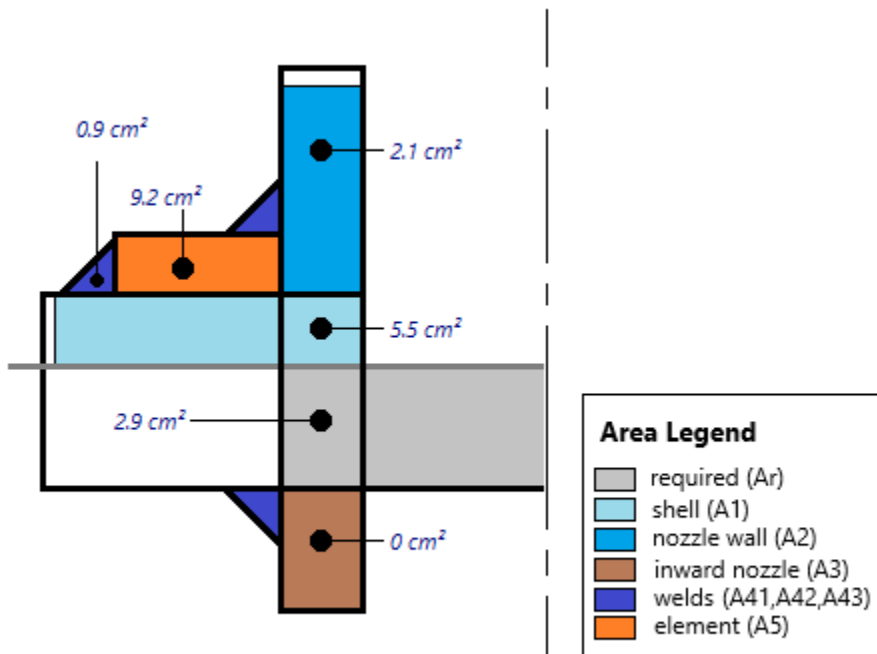


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2.169	2.929	NA
Area in Shell	A1	9.076	5.495	NA
Area in Nozzle Wall	A2	2.196	2.082	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.863	0.863	NA
Area in Element	A5	9.206	9.206	NA
TOTAL AREA AVAILABLE	Atot	21.341	17.646	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(103 \cdot 5.59 \cdot 1 + 2 \cdot 5.56 \cdot 5.59 \cdot 1(1 - 0.86))$$

$$= 2.929 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 103(1 \cdot 11 - 1 \cdot 5.59) - 2 \cdot 5.56$$

$$(1 \cdot 11 - 1 \cdot 5.59) \cdot (1 - 0.86)$$

$$= 5.495 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 23.9) \cdot (5.56 - 0.51) \cdot 0.86$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 153 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

$$= 2.082 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$$

$$= 0.863 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (206 - 114) \cdot 10 \cdot 1$$

$$= 9.206 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.5089 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 5.0715 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 3.3942 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.26, \max(5.5, 5.5)]$$

$$= 5.5000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.51, 5.5)$$

$$= 5.5000 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.560 = 7.490 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 38.7, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 256.3 N./mm² Passed

Occasional : 2.3, Allowable : 156.9 N./mm² Passed

Shear : 29.4, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 154 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

Govrn. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.19$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.23$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.052$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note:

The flange pressure ratio is less than 0.35. Impact testing is not required due to low stress per UHA-51(g).

Design Pressure/Ambient Rating = $530.39/1910.01 = 0.278$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds $T_{min} = 5.5598$ mm

Intermediate Calc. for pad/shell Welds $T_{minPad} = 10.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.8919 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2.93 - 5.49 + 2 \cdot 5.56 \cdot 0.86 \cdot (1 \cdot 11 - 5.59)) \cdot 137)$
 $= \max(0, -2855.85) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (W_i - C_{an} / 7.07)^2 \cdot f_r2) \cdot S_v$
 $= (2.08 + 9.21 + 0.86 - 0 \cdot 0.86) \cdot 137$
 $= 16950.00 \text{ Kgf}$

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (2.08 + 0 + 0.86 + (1.06)) \cdot 137$
 $= 5579.88 \text{ Kgf}$

Weld Load [W3]:

$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 155 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

$$= (2.08 + 0 + 0.86 + 9.21 + (1.06)) \cdot 137$$

$$= 18421.72 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 10585. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 214 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 23009. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 54.4) \cdot (8.56 - 3) \cdot 0.7 \cdot 118$$

$$= 7999. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 114 \cdot 10 \cdot 0.74 \cdot 137$$

$$= 18533. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 114 \cdot (14 - 3) \cdot 0.74 \cdot 137$$

$$= 20387. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (23009 + 7999) = 31007 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (10585 + 18533 + 20387 + 0) = 49505 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (23009 + 20387 + 0) = 43395 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 31007	Kgf, must exceed W = 0 Kgf	or W1 = 16949 Kgf
Path 2-2 = 49505	Kgf, must exceed W = 0 Kgf	or W2 = 5579 Kgf
Path 3-3 = 43395	Kgf, must exceed W = 0 Kgf	or W3 = 18421 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1148 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 3.0903 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 267.0903 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 37.21 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 103.18 mm

Input Echo, WRC297 Item 1, Description: Shell outlet :



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 156 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 136.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	97.180	mm
Nozzle Thickness	8.5598	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	214.300	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	530.39	KPa.
Radial Load	P	-765.00	Kgf
Circumferential Shear	Vc	510.00	Kgf
Longitudinal Shear	Vl	510.00	Kgf
Circumferential Moment	Mc	143.00	Kg-m.
Longitudinal Moment	Ml	143.00	Kg-m.
Torsional Moment	Mt	204.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{543.5(14.0 - 3.0)}$$

$$= 369.459 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1087.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	21.000	mm
Nozzle Thickness used	(t) =	5.560	mm
T / t	=	3.777	
d / t	=	20.558	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.757$$

$$Nr/P = 0.101$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 157 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9,2026

Mr/P = 0.098
MØ/P = 0.126
NØ/P = 0.265
MrD/Mc = 0.151
NrDL/Mc = 0.132
MØd/Mc = 0.238
NØDL/Mc = 0.122
MrD/Ml = 0.136
NrDL/Ml = 0.122
MØD/Ml = 0.209
NØDL/Ml = 0.251

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	13	13	13	13
Outplane Stress Summary	-11	33	72	-22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-7	17	44	-20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15	35	72	25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 158 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am Sep 9,2026

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	6	6	6	6

Outplane Stress Summary	-19	34	67	-38
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-25	25	25	-25
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	13	13	13	13

Inplane Stress Summary	-3	26	53	-16
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	19	35	69	39
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6

Hoop Stress Summary	3	3	17	17
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	69	-69	69	-69



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 159 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am Sep 9,2026

Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-28	-28	28	28
Axial Bending (ML)	-178	178	178	-178
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-130	87	282	-212
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	17	17	17	17
Shear Stress Summary	22	22	12	12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	141	94	283	231

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	6	7	13	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	-28	-28	28	28
Axial Bending (Mc)	-201	201	201	-201
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-153	110	305	-235

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	17	17	17	17
Shear Stress Summary	12	12	22	22

Nozzle Stresses (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 160 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9, 2026

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	162	111	307	253

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{538.5(14.0 - 3.0)}$$

$$= 368.282 \text{ mm}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 1077.000 \text{ mm}$$

$$\text{Nozzle Outside Diameter (d)} = 214.300 \text{ mm}$$

$$\text{Vessel Thickness used (T)} = 11.000 \text{ mm}$$

$$\text{Nozzle Thickness used (t)} = 55.560 \text{ mm}$$

$$T / t = 0.198$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.969$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.071
Mr/P	=	0.147
M0/P	=	0.043
N0/P	=	0.079
MrD/Mc	=	0.290
NrDL/Mc	=	0.089
M0d/Mc	=	0.089
N0DL/Mc	=	0.077
MrD/Ml	=	0.112
NrDL/Ml	=	0.088
M0D/Ml	=	0.035
N0DL/Ml	=	0.103

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-5	-5	5	5
Outplane Bending (Ml)	-11	11	11	-11
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	29	20	63	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-4	-4	4	4
Inplane Bending (Ml)	-36	36	36	-36



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 161 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am Sep 9,2026

Inplane Pressure Stress	12	12	12	12
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Inplane Stress Summary	30	-6	113	-69
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	34	27	113	78
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12

Outplane Stress Summary	0	26	66	-23
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-93	93	93	-93
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	25	25	25	25

Inplane Stress Summary	-14	64	183	-113
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 162 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9,2026

(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	15	64	183	113

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	13	13	13	13	13	13	13
Circ. Pl (SUS)	-2	-2	10	10	-2	-2	4	4
Circ. Q (SUS)	-22	22	46	-46	-16	16	34	-34
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-2	-2	4	4	1	1	7	7
Long. Q (SUS)	-13	13	31	-31	-27	27	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0
Pm+Pl (SUS)	11.1	11.1	23.1	23.1	11.2	11.2	17.2	17.2
Pm+Pl+Q (Total)	15.1	34.4	69.3	25.2	20.6	35.1	65.7	39.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	136.80	Passed
Pm+Pl (SUS)	23.08	205.20	Passed
Pm+Pl+Q (TOTAL)	69.32	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	25	25	25	25	25	25	25	25
Circ. Pl (SUS)	-1	-1	9	9	0	0	8	8
Circ. Q (SUS)	5	-5	27	-27	-39	39	147	-147
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	0	0	8	8	0	0	8	8
Long. Q (SUS)	18	-18	90	-90	-12	12	44	-44
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0
Pm+Pl (SUS)	24.1	24.1	34.1	34.1	25.1	25.1	33.1	33.1



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 163 of 214

Nozzle Calcs.: Shell outlet

Nozl: 15 10:11am

Sep 9,2026

Pm+Pl+Q (Total)	32.5	25.7	110.0	77.0	14.1	64.0	180.1	114.1
-----------------	------	------	-------	------	------	------	-------	-------

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	136.80	Passed
Pm+Pl (SUS)	34.07	205.20	Passed
Pm+Pl+Q (TOTAL)	180.08	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-2	-2	10	10	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-25	-25	31	31	-25	-25	31	31
Long. Q (SUS)	-109	109	247	-247	-132	132	270	-270
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	27.9	28.8	34.3	34.4	30.7	31.6	34.1	34.2
Pm+Pl+Q (Total)	142.0	93.1	280.5	231.2	162.8	110.4	304.7	253.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	34.36	177.00	Passed
Pm+Pl+Q (TOTAL)	304.65	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 164 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

Input, Nozzle Desc: Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	3340.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1331.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	136.80	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



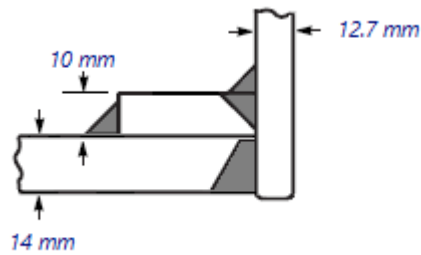
PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 165 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026



Reinforcement CALCULATION, Description: Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 520)$
 $= 2.0308 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 99.8 / (118 \cdot 1 - 0.6 \cdot 520)$
 $= 0.4412 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7465 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm
Parallel to Vessel Wall, opening length	d	199.6750 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.863$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.863$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.863$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 166 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

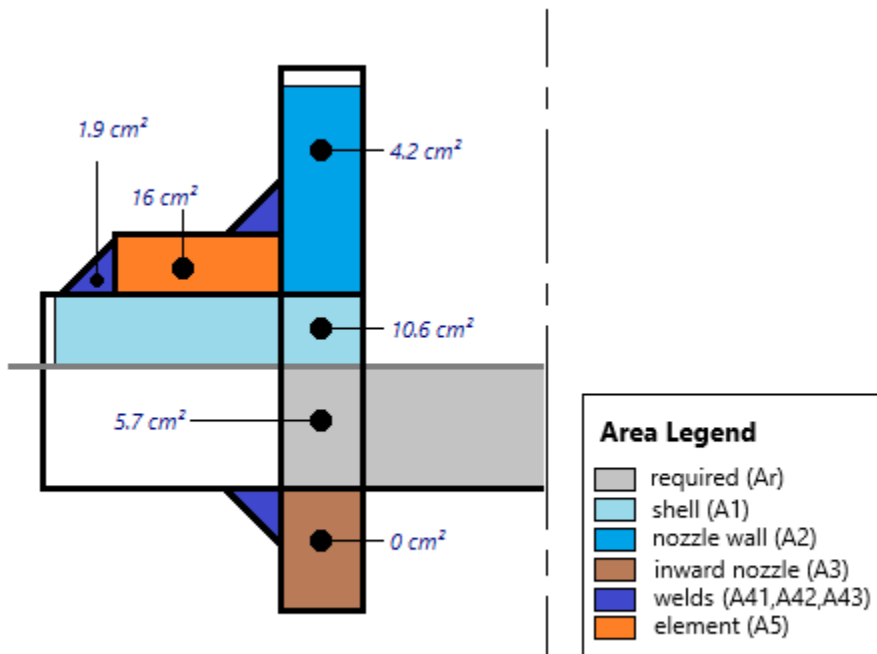


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4.109	5.660	NA
Area in Shell	A1	17.670	10.649	NA
Area in Nozzle Wall	A2	4.393	4.248	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.863	1.863	NA
Area in Element	A5	16.000	16.000	NA
TOTAL AREA AVAILABLE	Atot	39.925	32.759	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr_1)) \text{ per UG-37(d)}$$

$$= 0.5(200 \cdot 5.59 \cdot 1 + 2 \cdot 9.7 \cdot 5.59 \cdot 1(1-0.86))$$

$$= 5.660 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$$

$$= 200(1 \cdot 11 - 1 \cdot 5.59) - 2 \cdot 9.7$$

$$(1 \cdot 11 - 1 \cdot 5.59) \cdot (1 - 0.86)$$

$$= 10.649 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr_2$$

$$= (2 \cdot 27.5) \cdot (9.7 - 0.75) \cdot 0.86$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 167 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

= 4.248 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 10^2 \cdot 1$

= 1.863 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$

= $(379 - 219) \cdot 10 \cdot 1$

= 16.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.7465$ mm

Wall Thickness per UG16(b), $tr_{16b} = 5.5000$ mm

Wall Thickness, shell/head, internal pressure $trb_1 = 5.0308$ mm

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 5.5000$ mm

Wall Thickness, shell/head, external pressure $trb_2 = 3.3942$ mm

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000$ mm

Wall Thickness per table UG-45 $tb_3 = 10.1600$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[10.2, \max(5.5, 5.5)]$

= 5.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(3.75, 5.5)$

= 5.5000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 25.0, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 270.0 N./mm² Passed

Occasional : 2.6, Allowable : 156.9 N./mm² Passed

Shear : 17.8, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.1$, $t_r = 0.44$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.054$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 168 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 11.1$, $tr = 0.44$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.054$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $520.00 / 1960.01 = 0.265$

Weld Size Calculations, Description: Shell Inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 9.7000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5.66 - 10.6 + 2 \cdot 9.7 \cdot 0.86 \cdot (1 \cdot 11 - 5.59)) \cdot 137)$
 $= \max(0, -5697.76) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (4.25 + 16 + 1.86 - 0 \cdot 0.86) \cdot 137$
 $= 30842.20 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 169 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

$$= (4.25 + 0 + 0.86 + (1.84)) \cdot 137$$

$$= 9696.05 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (4.25 + 0 + 1.86 + 16 + (1.84)) \cdot 137$$

$$= 33409.87 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 20289. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 379 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 40700. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 105) \cdot (12.7 - 3) \cdot 0.7 \cdot 118$$

$$= 26870. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 219 \cdot 10 \cdot 0.74 \cdot 137$$

$$= 35522. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 219 \cdot (14 - 3) \cdot 0.74 \cdot 137$$

$$= 39074. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (40700 + 26870) = 67569 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (20289 + 35522 + 39074 + 0) = 94885 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (40700 + 39074 + 0) = 79774 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 67569	Kgf, must exceed W = 0 Kgf	or W1 = 30842 Kgf
Path 2-2 = 94884	Kgf, must exceed W = 0 Kgf	or W2 = 9696 Kgf
Path 3-3 = 79774	Kgf, must exceed W = 0 Kgf	or W3 = 33409 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2732 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 11.4428 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 275.4428 mm

Input Echo, WRC297 Item 1, Description: Shell Inlet :



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 170 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1060.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 136.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	379.075	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	520.00	KPa.
Radial Load	P	-1529.00	Kgf
Circumferential Shear	Vc	1019.00	Kgf
Longitudinal Shear	Vl	1019.00	Kgf
Circumferential Moment	Mc	571.00	Kg-m.
Longitudinal Moment	Ml	571.00	Kg-m.
Torsional Moment	Mt	816.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{543.5(14.0 - 3.0)}$$

$$= 474.234 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1087.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	21.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	2.165	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.450$$

Nr/P	=	0.084
Mr/P	=	0.096



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 171 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

MØ/P = 0.063
 NØ/P = 0.161
 MrD/Mc = 0.198
 NrDL/Mc = 0.133
 MØd/Mc = 0.148
 NØDL/Mc = 0.143
 MrD/Ml = 0.140
 NrDL/Ml = 0.094
 MØD/Ml = 0.100
 NØDL/Ml = 0.265

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	12	13	12	13
Outplane Stress Summary	-18	25	81	-13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	6	6	6	6
Inplane Stress Summary	-25	32	83	-53

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	29	36	85	53

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 172 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	6	6	6	6

Outplane Stress Summary	-35	42	84	-44
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-68	68	68	-68
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	12	13	12	13

Inplane Stress Summary	-40	57	111	-64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	42	58	113	66
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6

Hoop Stress Summary	-3	-3	26	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 173 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-151	151	151	-151
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-110	86	226	-180
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10

Shear Stress Summary	13	13	7	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	112	94	226	208
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6

Hoop Stress Summary	3	3	19	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-213	213	213	-213
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-173	149	288	-243
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	7	7	13	13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 174 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	177	150	289	265

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{538.5(14.0 - 3.0)}$$

$$= 473.057 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1077.000 mm
Nozzle Outside Diameter	(d) =	379.075 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	89.700 mm
T / t	=	0.123
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.483$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.056
Mr/P	=	0.094
M0/P	=	0.027
N0/P	=	0.042
MrD/Mc	=	0.260
NrDL/Mc	=	0.103
M0d/Mc	=	0.078
N0DL/Mc	=	0.067
MrD/Ml	=	0.070
NrDL/Ml	=	0.067
M0D/Ml	=	0.021
N0DL/Ml	=	0.050

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	24	25	24	25
Outplane Stress Summary	29	19	71	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	12	12	12	12



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 175 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am Sep 9,2026

Inplane Stress Summary	29	-7	149	-94
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	34	28	149	95
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-57	57	57	-57
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12

Outplane Stress Summary	-27	46	103	-51
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-190	190	190	-190
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	24	25	24	25

Inplane Stress Summary	-101	140	305	-215
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left	Left	Right	Right



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 176 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

	Outside	Inside	Outside	Inside
Two * Max Shear Stress	101	140	305	215

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	13	12	13	12	13	12	13
Circ. Pl (SUS)	-10	-10	20	20	-5	-5	9	9
Circ. Q (SUS)	-22	22	46	-46	-49	49	87	-87
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-3	-3	7	7	-3	-3	13	13
Long. Q (SUS)	-29	29	67	-67	-39	39	63	-63
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	12.0	13.0	12.0	13.0	12.0	13.0	12.0	13.0
Pm+Pl (SUS)	3.6	4.0	32.1	33.0	7.2	8.2	21.4	22.3
Pm+Pl+Q (Total)	29.7	35.4	83.1	54.4	44.0	58.0	109.3	66.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	136.80	Passed
Pm+Pl (SUS)	33.05	205.20	Passed
Pm+Pl+Q (TOTAL)	109.32	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	25	24	25	24	25	24	25
Circ. Pl (SUS)	-1	-1	11	11	-6	-6	18	18
Circ. Q (SUS)	5	-5	35	-35	-120	120	260	-260
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-2	-2	14	14	-3	-3	13	13
Long. Q (SUS)	19	-19	121	-121	-37	37	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	24.0	25.0	24.0	25.0	24.0	25.0	24.0	25.0
Pm+Pl (SUS)	23.1	24.1	35.1	36.1	18.1	19.1	42.1	43.1



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FileName : 192-C_BEM_Dia

Page 177 of 214

Nozzle Calcs.: Shell Inlet

Nozl: 16 10:11am

Sep 9, 2026

Pm+Pl+Q (Total)	32.5	29.1	147.1	96.1	102.1	139.0	302.1	217.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	136.80	Passed
Pm+Pl (SUS)	43.06	205.20	Passed
Pm+Pl+Q (TOTAL)	302.08	410.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-10	-10	20	20	-3	-3	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-15	-15	19	19	-15	-15	19	19
Long. Q (SUS)	-99	99	203	-203	-161	161	265	-265
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	12.5	12.5	26.8	26.8	15.5	15.5	23.0	23.0
Pm+Pl+Q (Total)	112.3	94.1	225.3	207.6	176.7	149.4	287.5	263.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	26.83	177.00	Passed
Pm+Pl+Q (TOTAL)	287.54	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 192-C_BEM_Dia

Page 178 of 214

Nozzle Calcs.: Vent

Nozl: 17 10:11am Sep 9,2026

Input, Nozzle Desc: Vent

From: 40

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	3340.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2981.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



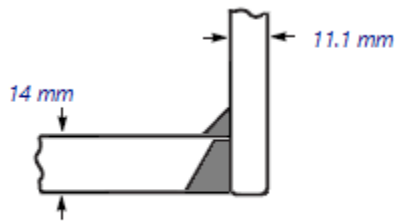
PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 179 of 214

Nozzle Calcs.: Vent

Nozl: 17 10:11am Sep 9,2026



Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 520)$
 $= 2.0308 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520 \cdot 22.1 / (118 \cdot 1 - 0.6 \cdot 520)$
 $= 0.0976 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3509 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524 mm
Parallel to Vessel Wall, opening length	d	44.1762 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.1860 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3509 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0308 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5000 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3942 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.5, 5.5)]$
 $= 5.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.35, 5.5)$
 $= 5.5000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 180 of 214

Nozzle Calcs.: Vent

Nozl: 17 10:11am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.098, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.015$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $520.00 / 1960.01 = 0.265$

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8590 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 264.8590 mm

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FileName : 192-C_BEM_Dia

Page 181 of 214

Nozzle Calcs.: Drain

Nozl: 18 10:11am Sep 9,2026

Input, Nozzle Desc: Drain

From: 40

Pressure for Reinforcement Calculations	P	530.390	KPa.
Temperature for Internal Pressure	Temp	280	°C
Design External Pressure	Pext	101.13	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	136.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1060.00	mm
Design Length of Section	L	3340.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2981.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0744	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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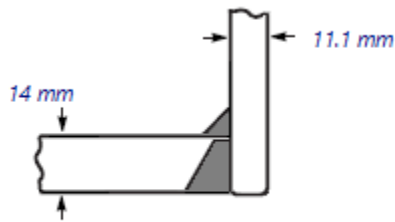
FileName : 192-C_BEM_Dia

Page 182 of 214

Nozzle Calcs.: Drain

Nozl: 18 10:11am

Sep 9, 2026



Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 530 \cdot 533 / (137 \cdot 1 - 0.6 \cdot 530)$
 $= 2.0715 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 530 \cdot 22.1 / (118 \cdot 1 - 0.6 \cdot 530)$
 $= 0.0996 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3509 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524 mm
Parallel to Vessel Wall, opening length	d	44.1762 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.1860 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3509 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.0715 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5000 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3942 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.5, 5.5)]$
 $= 5.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.35, 5.5)$
 $= 5.5000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$



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FileName : 192-C_BEM_Dia

Page 183 of 214

Nozzle Calcs.: Drain

Nozl: 18 10:11am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.1, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.015$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $530.39 / 1960.01 = 0.271$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds Tmin 8.0744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8590 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 264.8590 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 37.21 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 44.18 mm

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FileName : 192-C_BEM_Dia

Page 184 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	520.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1060.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	280.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	136.80 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell- left

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	900.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	900.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1060.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	895
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3340.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.15 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	510.398 mm



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FileName : 192-C_BEM_Dia

Page 185 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Largest Center-to-Center Tube Distance	U1	41.2750 mm
Length of Expanded Portion of Tube	ltx	40.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	132.50 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	80.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1280.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	1129.00 mm
Step 1 Depth on the Tubeside	ht1	6.00 mm
Step 2 Diameter on the Tubeside	dt2	1057.00 mm
Step 2 Depth on the Tubeside	ht2	5.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		1280.00 mm

Note: Tubesheet diameter is now: 1280.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	168.3 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	327.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	1280.000 mm
Flange Inside Diameter	B	1060.000 mm
Flange Face Outside Diameter	Fod	1129.000 mm
Flange Face Inside Diameter	Fid	1060.000 mm
Gasket Outside Diameter	Go	1118.000 mm
Gasket Inside Diameter	Gi	1092.000 mm
Small end Hub thk.	g0	14.0000 mm
Large end Hub thk.	g1	22.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	1190.000 mm
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FileName : 192-C_BEM_Dia

Page 186 of 214

ASME TS Calc: DESIGN 1

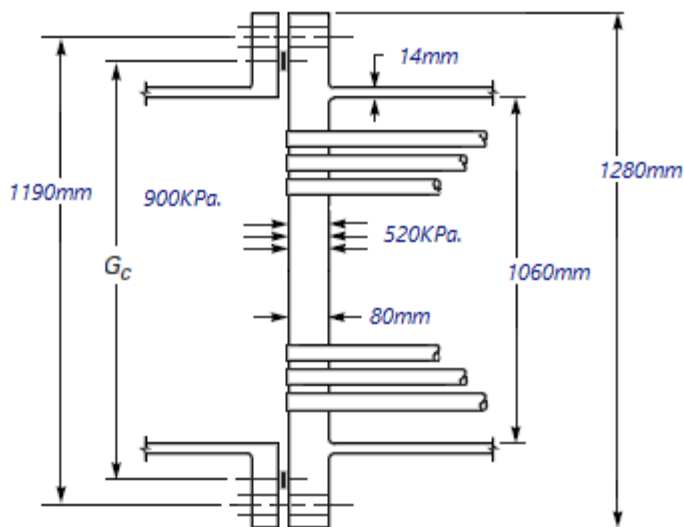
Case: 1 10:11a Sep 9, 2026

Nominal Bolt Diameter	dB	28.5750 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	36
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 44.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	280.0 °C	0.18624E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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FileName : 192-C_BEM_Dia

Page 187 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(900 \cdot (25.4/2 - 0)) / (92.1 \cdot 1 + 0.4 \cdot 900)$
= $0.1236 + 0.0000 = 0.1236$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 275.00 °C
Elastic Modulus for Material 183830896.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	3340.00	12.04	50.0000	0.0075908	80.31

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 80.3) / (3 \cdot 12) = 8894$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3984	25.40	3340.00	63.76	50.0000	0.0002706	24.87

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.9) / (3 \cdot 63.8) = 520$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3984 mm
Allowable Internal Pressure at Corroded thickness	16398.63 KPa.
Required Internal Design Pressure	900.00 KPa.
Allowable External Pressure at Corroded thickness	8894.21 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.3984 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	1105.156 mm

Required Tubesheet Flanged Extension thk. per 4.18.5 (operating) [hrop]:

= $\sqrt{1.9 \cdot WM1 \cdot hg / (S \cdot G)}$



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FileName : 192-C_BEM_Dia

Page 188 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= \sqrt{(1.9 \cdot 100318 \cdot 42.4 / (133 \cdot 1105))}$$

$$= 23.271 \text{ mm}$$

Required Tubesheet Flanged Extension thk. per 4.18.5 (seating) [hrseat]:

$$= \sqrt{(1.9 \cdot W \cdot h_g / (S_a \cdot G))}$$

$$= \sqrt{(1.9 \cdot 296547 \cdot 42.4 / (138 \cdot 1105))}$$

$$= 39.205 \text{ mm}$$

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 37/74$$

$$= 0.5000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(E_t/E), (St_t/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11(186735680/186735680) \cdot (92.1/133) \cdot (0.5); 25.4 - 2 \cdot 2.11)$$

$$= 23.9326 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 510 + 25.4$$

$$= 1046.196 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 1046/2$$

$$= 523.0980 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(168 \cdot 100; 4 \cdot 1046 \cdot 31.8) / (\pi \cdot 1046^2))}$$

$$= 32.0654 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.1 - 23.9) / 32.1$$

$$= 0.2536$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 533/523$$

$$= 1.0189$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 553/523$$

$$= 1.0564$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 895((25.4 - 2 \cdot 2.11) / (2 \cdot 523))^2$$

$$= 0.6332$$



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FileName : 192-C_BEM_Dia

Page 189 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 895 (25.4 / (2 \cdot 523))^2$$

$$= 0.4724$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((5 - 3); 0)$$

$$= 2.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 11 (1066 + 11)186239296/3346$$

$$= 211251.7031 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (25.4 - 2.11)186735680/3346$$

$$= 878.6172 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 211252 / (895 \cdot 879)$$

$$= 0.26864$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 211252/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1066 + 11)11)$$

$$= 0.0167 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.017 \cdot 186239296 \cdot 11^3 / (6(1 - 0.3^2))$$

$$= 758242752.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1066 + 11) \cdot 758242752 / 74^3 \cdot (1 + 74 \cdot 0.017 + (74 \cdot 0.017)^2 / 2)$$

$$= 36269936.0000 \text{ KPa}.$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1066^2 / (4 \cdot 186239296 \cdot 11) \cdot (1 - 1066 \cdot 0.3 / (2(1066 + 11)))$$

$$= 0.11808 \text{E+00 mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 2.330709$; $\mu^* = 0.253632$; $E^* = 41308316 \text{ KPa}$.

$E^*/E = 0.221213$; $\nu^* = 0.392756$

Note: As h/p (2.331) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$



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FileName : 192-C_BEM_Dia

Page 190 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= ((24(1-0.39^2)895 \cdot 186735680 \cdot 2.11 \\ (25.4 - 2.11)523^2) / (41308316 \cdot 3346 \cdot 74^3))^{1/4} \\ = 5.3421$$

Values from Table 4.18.3

$$Z_d = 0.009876 ; Z_v = 0.035023 ; Z_m = 0.268826 \\ Z_a = 0.359163E+02 ; Z_w = 0.035023$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o \\ = 1280/1046 \\ = 1.2235$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ = (1 - 0.39) / 41308316 \cdot (36269936 + 0 + \\ 186735680 \cdot \ln(1.22)) \\ = 1.0869$$

Compute parameter [ϕ]:

$$= (1 + v^*) F \\ = (1 + 0.39) 1.09 \\ = 1.5137$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ = (1.02 - 1 - 1.51 \cdot 0.035) / (1 + 1.51 \cdot 0.27) \\ = -0.024227$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ = (0.0099 + -0.024 \cdot 0.035) / 2 \cdot 5.34^4 \\ = 3.6761$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ = (0.035 + -0.024 \cdot 0.27) / 2 \cdot 5.34^4 \\ = 11.6099$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ = (0.035 + (1.02 - 1) 0.27) 5.34^4 / (1 + 1.51 \cdot 0.27) \\ = 23.2198$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)} \\ = (1105 - 1190) / 1046 = -0.081$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ = 1.02 \cdot 758242752 \cdot 0.017 \cdot 0.11807725 (1 + 74 \cdot 0.017) \\ = 3406.8076 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s \\ = 523^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 3407 \\ = -3357.3184 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 191 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Parameter [wc]:

$$= \rho c \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2))]$$

$$= 1.06 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 74) - 0.000 (0/74 + 0 \cdot 0/2))]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω*c]:

$$= ao^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c$$

$$= 523^2 [(1.06^2 + 1) \cdot (1.06 - 1) / 4 - (1.02 - 1) / 2] - 0$$

$$= 5567.3032 \text{ mm}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \}$$

$$= 0 \{ 0.47 + 2(1 - 0.47) 0.3 + [2/0.27 \cdot (1066/1046)^2] 0.3 - [(1.02^2 - 1) / (1 \cdot 0.27)] - [(1 - 1) / (2 \cdot 1 \cdot 0.27)] [(0^2 - (1066)^2) / 1046^2] \}$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt$$

$$= [0.63 + 2(1 - 0.63) 0.3 + 1 / (1 \cdot 0.27)] 900$$

$$= 4118.1021 \text{ KPa.}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 895 \cdot 879 \cdot 0 / (3.14 \cdot 523^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0.081 \cdot 23.2 \cdot 296547 / (2 \cdot \pi \cdot 523^2)$$

$$= 3185.0701 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= - (U / ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= - (23.2/523^2) \cdot (-3357 \cdot 0 - 5567 \cdot 900)$$

$$= 425.1858 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U / ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c)$$

$$= (23.2/523^2) \cdot (3407 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho s - 1) Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.27 / (1 + 1 \cdot 0.27 (3.68 + (1.02 - 1) 11.6)) \cdot (0 - 4118 + 0 + 0 + 3185 + 425)$$

$$= -66.6615 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega s \cdot P*s - \omega c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 192 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= [((-3357 \cdot 0 - 5567 \cdot 900) - (3407 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 296547 \cdot -0.081 / (2 \cdot \pi)] / (1 + 1.51 \cdot 0.27)$$

$$= -3083.6931 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.024 + 2 \cdot -3084 / (-66.7 \cdot 523^2)$$

$$= 3.29138$$

Fm Value from Table 4.18.3 = 1.658175

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 1.66 / 0.25) \cdot (2 \cdot 523 / (74 - 2))^2 \cdot -66.7$$

$$= -138.0319 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 133 = 199 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 1.47 / 0.25) \cdot (2 \cdot 523 / (61.6 - 2))^2 \cdot 74.2$$

$$= 198.7513 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 61.6 + 3 + 3$$

$$= 67.5970 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$$

$$= (2 \cdot 0.2 \cdot 74 / 523) \cdot 106$$

$$= 5.9982 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (1 / (2 \cdot \mu)) \cdot (ao / h) \cdot Pe$$

$$= (1 / (2 \cdot 0.2)) \cdot (523 / 12.1) \cdot 968$$

$$= 104.5256 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + Cats + Catc$$

$$= 12.1 + 3 + 3$$

$$= 18.1127 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(HreqB; HreqS)$$

$$= \max(67.6; 18.1)$$

$$= 67.5970 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -15.3, F_{tmax} = 51$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs)$$

$$= ((0 \cdot 0.47 - 900 \cdot 0.63) - (-66.7) \cdot -15.3) / (0.63 - 0.47)$$

$$= -9.8967 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 193 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((0 \cdot 0.47 - 900 \cdot 0.63) - (-66.7) \cdot 51) / (0.63 - 0.47)$$

$$= 17.5917 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 17.6 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.1486 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.0099 + 3.29 \cdot 0.035) 5.34^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 186735680 / 144)}$$

$$= 159.9957$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 327 / 8.27$$

$$= 39.5502$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (144 / 1.25) \cdot (1 - 39.6 / (2 \cdot 160))$$

$$= 92.149 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 17.6 \cdot 1.54$$

$$= 276.94 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 92.1$$

$$= 1450.647 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$$

$$= 1450.6471 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 194 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= 92.1/92.1$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$$

$$= 1335.0663 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1451 is $\geq$ tube strength 1451 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 137 \cdot 1 = 137 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 523^2 / ((1066 + 11) 11) \cdot [-66.7 + (1.02^2 - 1)(0 - 900)] + 533^2 \cdot 900 / ((1066 + 11) 11)$$

$$= 19.2488 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 11 mm, the minimum Shell length adjacent to the tubesheet should be 194.916 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 523^2 / ((1066 + 11) 11) [-66.7 + (1.02^2 - 1)(0 - 900)] + 533^2 \cdot 900 / ((1066 + 11) 11)$$

$$= 19.2488 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 758242752 / 11^2 \{ 0.017 [0.11807725 \cdot 0 + 533^2 \cdot 0 / (2048632320)] + 6(1 - 0.39^2) / (41308316) (523/74)^3 (1 + 74 \cdot 0.017/2) [-66.7 (0.035 + 0.27 \cdot -0.024) + 2/523^2 \cdot 0.27 \cdot -3084] \}$$

$$= -161.8651 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s, m}| + |\sigma_{s, b}| \leq 1.5 \cdot S_s$$

$$\therefore |19.2| + |-162| \leq 205 \text{ N./mm}^2$$

$$\therefore 181, \text{ must be } \leq 205 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 195 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

Stress/Force Summary for Loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	138.0 <=	198.8 N./mm ²	Pass
Tubesheet Shear Stress	1.2 <=	106.0 N./mm ²	Pass
Maximum Tube Stress	17.6 <=	92.1 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-9.9 <=	-92.1 N./mm ²	Pass
Maximum Force on any one Tube	276.9 <=	1450.6 Kgf	Pass
Axial Membrane Stress in Shell	19.2 <=	136.8 N./mm ²	Pass
Shell Stress (jt. inter.)	181.1 <=	205.2 N./mm ²	Pass

Thickness Results for Loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	67.597	80.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	1105.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

= lt_x / h (modified for corrosion if present)

= 37/74

= 0.5000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(25.4 - 2 \cdot 2.11 (186735680/186735680) \cdot$

$(92.1/133) \cdot (0.5); 25.4 - 2 \cdot 2.11)$

= 23.9326 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$

= $2 \cdot 510 + 25.4$

= 1046.196 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= 1046/2



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 196 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

= 523.0980 mm

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(168 \cdot 100 ; 4 \cdot 1046 \cdot 31.8) / (\pi \cdot 1046^2))}$$

$$= 32.0654 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.1 - 23.9) / 32.1$$

$$= 0.2536$$

Ratio [ps]:

$$= as / ao$$

$$= 533 / 523$$

$$= 1.0189$$

Ratio [pc]:

$$= ac / ao$$

$$= 553 / 523$$

$$= 1.0564$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 895 ((25.4 - 2 \cdot 2.11) / (2 \cdot 523))^2$$

$$= 0.6332$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 895 (25.4 / (2 \cdot 523))^2$$

$$= 0.4724$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((5 - 3); 0)$$

$$= 2.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts) Es / L$$

$$= \pi \cdot 11 (1066 + 11) 186239296 / 3346$$

$$= 211251.7031 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt) Et / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 186735680 / 3346$$

$$= 878.6172 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 211252 / (895 \cdot 879)$$

$$= 0.26864$$

Rigidty Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 211252 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1066 + 11) 11)$$

$$= 0.0167 \text{ 1/mm}$$

Shell coefficient [ks]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 197 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.017 \cdot 186239296 \cdot 11^3 / (6(1 - 0.3^2)) \\
 &= 758242752.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1066 + 11) \cdot 758242752 / 74^3 \cdot (1 + 74 \cdot 0.017 + (74 \cdot 0.017)^2 / 2) \\
 &= 36269936.0000 \text{ KPa}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\
 &= 1066^2 / (4 \cdot 186239296 \cdot 11) \cdot (1 - 1066 \cdot 0.3 / (2(1066 + 11))) \\
 &= 0.11808\text{E}+00 \text{ mm/N./mm}^2
 \end{aligned}$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned}
 h/p &= 2.330709 ; \mu^* = 0.253632 ; E^* = 41308316 \text{ KPa} \\
 E^*/E &= 0.221213 ; \nu^* = 0.392756
 \end{aligned}$$

Note: As h/p (2.331) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$\begin{aligned}
 &= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4} \\
 &= ((24(1 - 0.39^2)895 \cdot 186735680 \cdot 2.11 \\
 &\quad (25.4 - 2.11)^{5/2}) / (41308316 \cdot 3346 \cdot 74^3))^{1/4} \\
 &= 5.3421
 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned}
 Z_d &= 0.009876 ; Z_v = 0.035023 ; Z_m = 0.268826 \\
 Z_a &= 0.359163\text{E}+02 ; Z_w = 0.035023
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 1280 / 1046 \\
 &= 1.2235
 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.39) / 41308316 \cdot (36269936 + 0 + \\
 &\quad 186735680 \cdot \ln(1.22)) \\
 &= 1.0869
 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned}
 &= (1 + \nu^*)F \\
 &= (1 + 0.39)1.09 \\
 &= 1.5137
 \end{aligned}$$

Compute parameter [Q_1]:

$$\begin{aligned}
 &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\
 &= (1.02 - 1 - 1.51 \cdot 0.035) / (1 + 1.51 \cdot 0.27) \\
 &= -0.024227
 \end{aligned}$$

Compute Parameter [Q_{z1}]:

$$\begin{aligned}
 &= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4 \\
 &= (0.0099 + -0.024 \cdot 0.035) / 2 \cdot 5.34^4 \\
 &= 3.6761
 \end{aligned}$$

Compute parameter [Q_{z2}]:

$$\begin{aligned}
 &= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4 \\
 &= (0.035 + -0.024 \cdot 0.27) / 2 \cdot 5.34^4
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 198 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

= 11.6099

Compute parameter [U]:

$$= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.035 + (1.02 - 1)0.27)5.34^4 / (1 + 1.51 \cdot 0.27)$$

$$= 23.2198$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= (Gc - C) / Do \text{ (config b)}$$

$$= (1105 - 1190) / 1046 = -0.081$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.02 \cdot 758242752 \cdot 0.017 \cdot 0.11807725 (1 + 74 \cdot 0.017)$$

$$= 3406.8076 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - ws$$

$$= 523^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 3407$$

$$= -3357.3184 \text{ mm}^2$$

Parameter [wc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.06 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 74) -$$

$$0.000 (0/74 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [w*c]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc$$

$$= 523^2 [(1.06^2 + 1) \cdot (1.06 - 1) / 4 - (1.02 - 1) / 2] - 0$$

$$= 5567.3032 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs -$$

$$[(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 520 \{ 0.47 + 2(1 - 0.47)0.3 + [2/0.27 \cdot$$

$$(1066/1046)^2]0.3 - [(1.02^2 - 1)/(1 \cdot 0.27)] -$$

$$[(1 - 1)/(2 \cdot 1 \cdot 0.27)] [(0^2 - (1066)^2)/1046^2] \}$$

$$= 1542.0665 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] P't$$

$$= [0.63 + 2(1 - 0.63)0.3 + 1/(1 \cdot 0.27)] 0$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pv]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 895 \cdot 879 \cdot 0 / (3.14 \cdot 523^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 199 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$\begin{aligned}
 &= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\
 &= -0.081 \cdot 23.2 \cdot 0 / (2 \cdot \pi \cdot 523^2) \\
 &= 0.0000 \text{ KPa.}
 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned}
 &= -(U/a_o^2) \cdot (\omega^*s \cdot P_s - \omega^*c \cdot P_t) \\
 &= -(23.2/523^2) \cdot (-3357 \cdot 520 - 5567 \cdot 0) \\
 &= 148.1451 \text{ KPa.}
 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned}
 &= (U/a_o^2) \cdot (\omega_s \cdot P^*s - \omega_c \cdot P^*c) \\
 &= (23.2/523^2) \cdot (3407 \cdot 0 - 0 \cdot 0) \\
 &= 0.0000 \text{ KPa.}
 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned}
 &= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot \\
 &\quad (P^*s - P^*t + P_\gamma + P_\omega + P_w + P_{\text{Prim}}) \\
 &= 0.1000E+01 \cdot 0.27 / (1 + 1 \cdot 0.27 (3.68 + (1.02 - 1) 11.6)) \cdot (1542 - 0 + 0 + 0 + 0 + 148) \\
 &= 221.8627 \text{ KPa.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega^*s \cdot P_s - \omega^*c \cdot P_t) - (\omega_s \cdot P^*s - \omega_c \cdot P^*c)) + \\
 &\quad W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((-3357 \cdot 520 - 5567 \cdot 0) - \\
 &\quad (3407 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\
 &\quad 0 \cdot -0.081 / (2 \cdot \pi)] / (1 + 1.51 \cdot 0.27) \\
 &= -126.5379 \text{ Kg}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2) \\
 &= -0.024 + 2 \cdot -127 / (222 \cdot 523^2) \\
 &= -0.06511
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.032553

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\
 &= (1.5 \cdot 0.033 / 0.25) \cdot (2 \cdot 523 / (74 - 2))^2 \cdot 222 \\
 &= 9.0188 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$1.5 \cdot S = 1.5 \cdot 133 = 199 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e \\
 &= (1.5 \cdot 0.016 / 0.25) \cdot (2 \cdot 523 / (10.2 - 2))^2 \cdot 130 \\
 &= 198.7352 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 10.2 + 3 + 3 \\
 &= 16.1876 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y) \\
 &= (2 \cdot 0.2 \cdot 74 / 523) \cdot 106 \\
 &= 5.9982 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ_{limit}



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 200 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 9.8100 mm

Tubesheet Shear Stress = 28.9760 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

= max(HreqB; HreqS)

= max(16.2 ; 9.81)

= 16.1876 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

Ftmin = -0.42, Ftmax = 3.09

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

= ((Ps • xs - Pt • xt) - Pe • Ftmin) / (Xt - Xs)

= ((520 • 0.47 - 0 • 0.63) - (222) • -0.42) /

(0.63 - 0.47)

= 2.1024 N./mm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

= ((Ps • xs - Pt • xt) - Pe • Ftmax) / (Xt - Xs)

= ((520 • 0.47 - 0 • 0.63) - (222) • 3.09) /

(0.63 - 0.47)

= -2.7411 N./mm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

= max(| $\sigma_{t,1}$ | ; | $\sigma_{t,2}$ |) = 2.74 N./mm²

Allowable Tube Stress [$\sigma_{t,A}$]:

= Sot = 92.1486 N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

= max((3.25 - 0.25(Zd + Q3 • Zw)Xa⁴); 1.25)

= max((3.25 - 0.25(0.0099 + -0.065 • 0.035)5.34⁴); 1.25)

= 1.7035 (Should be ≤ 2)

Determine the factor [rt]:

= ($\sqrt{dt^2 + (dt - 2 \cdot tt)^2}$) / 4

= ($\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}$) / 4

= 8.2680 mm

Determine the factor [Ct]:

= $\sqrt{(2 \cdot \pi^2 \cdot Et / Syt)}$

= $\sqrt{(2 \cdot \pi^2 \cdot 186735680 / 144)}$

= 159.9957

Determine the factor [Ft]:

= k • L/r

= 1 • 327/8.27

= 39.5502

Buckling Allowable Stress [Stb]:

= (Sy,t/Fs) • (1 - Ft/(2 • Ct))

= (144/1.7) • (1 - 39.6/(2 • 160))

= 74.086 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

= $\sigma_{t,max}$ • Tube Area

= 2.74 • 1.54



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 201 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

= 43.15 Kgf

Tube Metal Area [At]:

= $\pi (d_o - t) t$

= $\pi (25.4 - 2.11) 2.11$

= 1.544 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

= $\pi \cdot t (d_o - t) S$

= $\pi \cdot 2.11 (25.4 - 2.11) 92.1$

= 1450.647 Kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)

= $0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$

= 1450.6471 Kgf

Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$

= 92.1/92.1

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)

= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$

= 1335.0663 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1451 is >= tube strength 1451 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

= $S_s \cdot E_{sw} = 137 \cdot 1 = 137 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:

= $a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$

= $523^2 / ((1066 + 11) 11) \cdot [222 + (1.02^2 - 1)$

$(520 - 0)] + 533^2 \cdot 0 / ((1066 + 11) 11)$

= 5.5837 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 11 mm, the minimum Shell length adjacent to the tubesheet should be 194.916 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

= $a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 202 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

$$= 523^2 / ((1066 + 11)11) [222 + (1.02^2 - 1)(520 - 0)] + 533^2 \cdot 0 / ((1066 + 11)11)$$

$$= 5.5837 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P_s / (E_s \cdot t_s)] + 6(1 - \nu_s^2) / (E_s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 758242752 / 11^2 \{ 0.017 [0.11807725 \cdot 520 + 533^2 \cdot 0 / (2048632320)] + 6(1 - 0.39^2) / (41308316) (523 / 74)^3 (1 + 74 \cdot 0.017 / 2) [222 (0.035 + 0.27 \cdot -0.024) + 2 / 523^2 \cdot 0.27 \cdot -127] \}$$

$$= 48.8221 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$

$$:: |5.58| + |48.8| \leq 205 \text{ N./mm}^2$$

$$:: 54.4, \text{ must be } \leq 205 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	9.0 <=	198.8 N./mm ²	Pass
Tubesheet Shear Stress	3.9 <=	106.0 N./mm ²	Pass
Maximum Tube Stress	2.7 <=	92.1 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.7 <=	-74.1 N./mm ²	Pass
Maximum Force on any one Tube	43.2 <=	1450.6 Kgf	Pass
Axial Membrane Stress in Shell	5.6 <=	136.8 N./mm ²	Pass
Shell Stress (jt. inter.)	54.4 <=	205.2 N./mm ²	Pass

Thickness Results for Loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	16.188	80.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	1105.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 37 / 74$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 203 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= 0.5000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt) \\ &= \max(25.4 - 2 \cdot 2.11 (186735680/186735680) \cdot \\ &\quad (92.1/133) \cdot (0.5); 25.4 - 2 \cdot 2.11) \\ &= 23.9326 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 510 + 25.4 \\ &= 1046.196 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1046/2 \\ &= 523.0980 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(168 \cdot 100 ; 4 \cdot 1046 \cdot 31.8) / (\pi \cdot 1046^2)) } \\ &= 32.0654 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ*]:

$$\begin{aligned} &= (p* - d*) / p* \\ &= (32.1 - 23.9) / 32.1 \\ &= 0.2536 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 533/523 \\ &= 1.0189 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 553/523 \\ &= 1.0564 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 895 ((25.4 - 2 \cdot 2.11) / (2 \cdot 523))^2 \\ &= 0.6332 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 895 (25.4 / (2 \cdot 523))^2 \\ &= 0.4724 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((5 - 3); 0) \\ &= 2.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(Ds + ts)Es/L \\ &= \pi \cdot 11 (1066 + 11)186239296/3346 \\ &= 211251.7031 \text{ Kg/mm} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 204 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t \cdot (D_t - t) \cdot E / L$$

$$= \pi \cdot 2.11 \cdot (25.4 - 2.11) \cdot 186735680 / 3346$$

$$= 878.6172 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 211252 / (895 \cdot 879)$$

$$= 0.26864$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 211252 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1066 + 11) \cdot 11)$$

$$= 0.0167 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.017 \cdot 186239296 \cdot 11^3 / (6(1 - 0.3^2))$$

$$= 758242752.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1066 + 11) \cdot 758242752 / 74^3 \cdot (1 + 74 \cdot 0.017 + (74 \cdot 0.017)^2 / 2)$$

$$= 36269936.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1066^2 / (4 \cdot 186239296 \cdot 11) \cdot (1 - 1066 \cdot 0.3 / (2(1066 + 11)))$$

$$= 0.11808 \text{E+00 mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 2.330709$; $\mu^* = 0.253632$; $E^* = 41308316 \text{ KPa}$.

$E^*/E = 0.221213$; $\nu^* = 0.392756$

Note: As h/p (2.331) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2}) N_t \cdot E_t \cdot t \cdot (d_t - t) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.39^2) 895 \cdot 186735680 \cdot 2.11 \cdot (25.4 - 2.11)^2) / (41308316 \cdot 3346 \cdot 74^3))^{1/4}$$

$$= 5.3421$$

Values from Table 4.18.3

$Z_d = 0.009876$; $Z_v = 0.035023$; $Z_m = 0.268826$
 $Z_a = 0.359163 \text{E+02}$; $Z_w = 0.035023$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1280 / 1046$$

$$= 1.2235$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.39) / 41308316 \cdot (36269936 + 0 +$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 205 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$186735680 \cdot \ln(1.22) \\ = 1.0869$$

Compute parameter [ϕ]:

$$= (1 + \nu)F \\ = (1 + 0.39)1.09 \\ = 1.5137$$

Compute parameter [Q1]:

$$= (\psi - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ = (1.02 - 1 - 1.51 \cdot 0.035) / (1 + 1.51 \cdot 0.27) \\ = -0.024227$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ = (0.0099 + -0.024 \cdot 0.035) / 2 \cdot 5.34^4 \\ = 3.6761$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ = (0.035 + -0.024 \cdot 0.27) / 2 \cdot 5.34^4 \\ = 11.6099$$

Compute parameter [U]:

$$= (Z_w + (\psi - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m) \\ = (0.035 + (1.02 - 1)0.27)5.34^4 / (1 + 1.51 \cdot 0.27) \\ = 23.2198$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)} \\ = (1105 - 1190) / 1046 = -0.081$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \psi \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ = 1.02 \cdot 758242752 \cdot 0.017 \cdot 0.11807725 (1 + 74 \cdot 0.017) \\ = 3406.8076 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (\psi^2 - 1) \cdot (\psi - 1) / 4 - \omega_s \\ = 523^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 3407 \\ = -3357.3184 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)] \\ = 1.06 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 74) - \\ 0.000 (0/74 + 0 \cdot 0/2)] \\ = 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\psi - 1) / 2] - \omega_c \\ = 523^2 [(1.06^2 + 1) \cdot (1.06 - 1) / 4 - (1.02 - 1) / 2] - 0 \\ = 5567.3032 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 206 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned}
 &= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] \} \text{ vs } - \\
 &\quad [(\rho s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \} \\
 &= 520 \{ 0.47 + 2(1 - 0.47) 0.3 + [2/0.27 \cdot \\
 &\quad (1066/1046)^2] 0.3 - [(1.02^2 - 1)/(1 \cdot 0.27)] - \\
 &\quad [(1 - 1)/(2 \cdot 1 \cdot 0.27)] [(0^2 - (1066)^2)/1046^2] \} \\
 &= 1542.0665 \text{ KPa.}
 \end{aligned}$$

Pressure [P't]:

$$\begin{aligned}
 &= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t \\
 &= [0.63 + 2(1 - 0.63) 0.3 + 1/(1 \cdot 0.27)] 900 \\
 &= 4118.1021 \text{ KPa.}
 \end{aligned}$$

Pressure [P_y]:

$$\begin{aligned}
 &= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\
 &= 895 \cdot 879 \cdot 0 / (3.14 \cdot 523^2) \\
 &= 0.0000 \text{ KPa.}
 \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned}
 &= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\
 &= -0.081 \cdot 23.2 \cdot 296547 / (2 \cdot \pi \cdot 523^2) \\
 &= 3185.0701 \text{ KPa.}
 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned}
 &= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t) \\
 &= -(23.2/523^2) \cdot (-3357 \cdot 520 - 5567 \cdot 900) \\
 &= 573.3310 \text{ KPa.}
 \end{aligned}$$

Calculate the pressure [P_w]:

$$\begin{aligned}
 &= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c) \\
 &= (23.2/523^2) \cdot (3407 \cdot 0 - 0 \cdot 0) \\
 &= 0.0000 \text{ KPa.}
 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned}
 &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (\rho s - 1)Qz2)) \cdot \\
 &\quad (P's - P't + P_y + P_w + P_w + Prim) \\
 &= 0.1000E+01 \cdot 0.27 / (1 + 1 \cdot 0.27(3.68 + (1.02 - \\
 &\quad 1)11.6)) \cdot (1542 - 4118 + 0 + 0 + 3185 + 573) \\
 &= 155.2012 \text{ KPa.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega_s \cdot P^* s - \omega_c \cdot P^* c)) + \\
 &\quad W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((-3357 \cdot 520 - 5567 \cdot 900) - \\
 &\quad (3407 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\
 &\quad 296547 \cdot -0.081 / (2 \cdot \pi)] / (1 + 1.51 \cdot 0.27) \\
 &= -3210.2307 \text{ Kgf}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2) \\
 &= -0.024 + 2 \cdot -3210 / (155 \cdot 523^2) \\
 &= -1.50677
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.753387

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.75/0.25) \cdot (2 \cdot 523 / (74 - 2))^2 \cdot 155 \\
 &= 146.0115 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 207 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

$$= 1.5 \cdot S = 1.5 \cdot 133 = 199 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.44 / 0.25) \cdot (2 \cdot 523 / (63.4 - 2))^2 \cdot 264$$

$$= 198.7503 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 63.4 + 3 + 3$$

$$= 69.3576 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 74 / 523) \cdot 106$$

$$= 5.9982 \text{ N./mm}^2$$

Shear Stress is not required to be computed; $[P_e] \leq T_{aulimit}$

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (1 / (2 \cdot \mu)) \cdot (a_o / h) \cdot P_e$$

$$= (1 / (2 \cdot 0.2)) \cdot (523 / 14) \cdot 1117$$

$$= 104.1736 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 14 + 3 + 3$$

$$= 20.0285 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(69.4; 20)$$

$$= 69.3576 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -17.5, F_{tmax} = 6.81$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((520 \cdot 0.47 - 900 \cdot 0.63) - (155) \cdot -17.5) /$$

$$(0.63 - 0.47)$$

$$= 14.8506 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((520 \cdot 0.47 - 900 \cdot 0.63) - (155) \cdot 6.81) /$$

$$(0.63 - 0.47)$$

$$= -8.5975 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 14.9 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.1486 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.0099 + -1.51 \cdot 0.035)5.34^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 208 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Determine the factor [rt]:

$$= \left(\sqrt{\left(\frac{dt^2}{2} + (dt - 2 \cdot tt)^2 \right)} \right) / 4$$

$$= \left(\sqrt{\left(\frac{25.4^2}{2} + (25.4 - 2 \cdot 2.11)^2 \right)} \right) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot Et}{Syt} \right)}$$

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot 186735680}{144} \right)}$$

$$= 159.9957$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 1 \cdot 327/8.27$$

$$= 39.5502$$

Buckling Allowable Stress [Stb]:

$$= (Sy, t/Fs) \cdot (1 - Ft/(2 \cdot Ct))$$

$$= (144/2) \cdot (1 - 39.6/(2 \cdot 160))$$

$$= 63.101 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t, \max} \cdot \text{Tube Area}$$

$$= 14.9 \cdot 1.54$$

$$= 233.79 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 92.1$$

$$= 1450.647 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$$

$$= 1450.6471 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 92.1/92.1$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.1$$

$$= 1335.0663 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot do}{2} \right)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot 25.4}{2} \right)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm



PV Elite 28 Licensee: SL User

FileName : 192-C_BEM_Dia

Page 209 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1451 is >= tube strength 1451 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 137 \cdot 1 = 137 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1) (P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 523^2 / ((1066 + 11) 11) \cdot [155 + (1.02^2 - 1) (520 - 900)] + 533^2 \cdot 900 / ((1066 + 11) 11)$$

$$= 24.8326 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 11 mm, the minimum Shell length adjacent to the tubesheet should be 194.916 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 523^2 / ((1066 + 11) 11) [155 + (1.02^2 - 1) (520 - 900)] + 533^2 \cdot 900 / ((1066 + 11) 11)$$

$$= 24.8326 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E \cdot t_s)] + 6 (1 - \nu^2) / (E \cdot h) (a o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 758242752 / 11^2 \{ 0.017 [0.11807725 \cdot 520 + 533^2 \cdot 0 / (2048632320)] + 6 (1 - 0.39^2) / (41308316) (523 / 74)^3 (1 + 74 \cdot 0.017 / 2) [155 (0.035 + 0.27 \cdot -0.024) + 2 / 523^2 \cdot 0.27 \cdot -3210] \}$$

$$= -113.0430 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore | \sigma_{s, m} | + | \sigma_{s, b} | \leq 1.5 \cdot S_s$$

$$\therefore | 24.8 | + | -113 | \leq 205 \text{ N./mm}^2$$

$$\therefore 138, \text{ must be } \leq 205 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	146.0 <=	198.8 N./mm ²	Pass
Tubesheet Shear Stress	2.7 <=	106.0 N./mm ²	Pass
Maximum Tube Stress	14.9 <=	92.1 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-8.6 <=	-63.1 N./mm ²	Pass
Maximum Force on any one Tube	233.8 <=	1450.6 Kgf	Pass
Axial Membrane Stress in Shell	24.8 <=	136.8 N./mm ²	Pass
Shell Stress (jt. inter.)	137.9 <=	205.2 N./mm ²	Pass

Thickness Results for Loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	69.358	80.000	Ok



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FileName : 192-C_BEM_Dia

Page 210 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

Tube-Tubesheet Fillet Weld Leg :	1.333	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Tbsht	Thk. + CA Extnsn	---- Tubesheet Bend	Stresses Allwd	Shear Allwd	Case Type	Pass/Fail
D1uc	62.140	39.205	120	199	1	106 Ps+Pt-Th	D1 Ok
D2uc	15.058	...	11	199	4	106 Ps+Pt-Th	D2 Ok
D3uc	64.601	...	129	199	2	106 Ps+Pt-Th	D3 Ok
O1uc	37.175	...	105	398	1	106 Ps+Pt+Th	O1 Ok
O2uc	37.432	...	109	398	10	106 Ps+Pt+Th	O2 Ok
O3uc	38.904	...	114	398	2	106 Ps+Pt+Th	O3 Ok
O4uc	35.662	...	99	398	6	106 Ps+Pt+Th	O4 Ok
D1c	67.597	39.205	138	199	1	106 Ps+Pt-Th-c	D1 Ok
D2c	16.188	...	9	199	4	106 Ps+Pt-Th-c	D2 Ok
D3c	69.358	...	146	199	3	106 Ps+Pt-Th-c	D3 Ok
O1c	46.712	...	131	398	1	106 Ps+Pt+Th-c	O1 Ok
O2c	45.921	...	130	398	11	106 Ps+Pt+Th-c	O2 Ok
O3c	47.837	...	138	398	3	106 Ps+Pt+Th-c	O3 Ok
O4c	44.754	...	121	398	7	106 Ps+Pt+Th-c	O4 Ok
Max:	69.3576	39.205 mm	0.735	0.107	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Ten	Shell Stresses Allwd	Cmp	---- Allwd	:	---- Ten	Shell Band Stress Allwd	Cmp	---- Allwd	: Pass/Fail
D1uc	15	136	...	...	:	...	...	...	...	Ok
D2uc	4	136	...	...	:	...	...	...	...	Ok
D3uc	19	136	...	...	:	...	...	...	...	Ok
O1uc	15	417	...	...	:	...	...	...	...	Ok
O2uc	12	417	...	...	:	...	...	...	...	Ok
O3uc	19	417	...	...	:	...	...	...	...	Ok
O4uc	7	417	...	...	:	...	...	...	...	Ok
D1c	19	136	...	...	:	...	...	...	...	Ok
D2c	6	136	...	...	:	...	...	...	...	Ok



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FileName : 192-C_BEM_Dia

Page 211 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

D3c	25	136	...	...	...	...	...	...	Ok
01c	19	417	...	...	...	...	...	...	Ok
02c	15	417	...	...	...	...	...	...	Ok
03c	25	417	...	...	...	...	...	...	Ok
04c	10	417	...	...	...	...	...	...	Ok

Max RATIO 0.182 ...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel	Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	15	92	-9	-92	229	1450	158	205	...	...	Ok
D2uc	2	92	-2	-67	37	1450	43	205	...	...	Ok
D3uc	12	92	-8	-63	192	1450	125	205	...	...	Ok
01uc	15	184	-9	-92	229	2901	158	417	...	...	Ok
02uc	5	184	-5	-63	76	2901	67	417	...	...	Ok
03uc	12	184	-8	-63	192	2901	125	417	...	...	Ok
04uc	6	184	-5	-63	97	2901	101	417	...	...	Ok
D1c	18	92	-10	-92	277	1450	181	205	...	...	Ok
D2c	3	92	-3	-74	43	1450	54	205	...	...	Ok
D3c	15	92	-9	-63	234	1450	138	205	...	...	Ok
01c	18	184	-10	-92	277	2901	181	417	...	...	Ok
02c	5	184	-5	-63	86	2901	73	417	...	...	Ok
03c	15	184	-9	-63	234	2901	138	417	...	...	Ok
04c	8	184	-6	-63	122	2901	117	417	...	...	Ok

Max RATIO 0.191 0.136 0.191 0.883 ...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	69.358	80.000	Ok
Tubesheet thickness flanged extension :	39.205	44.000	Ok
Tube thickness :	0.398	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	3.000	Ok
Tube-tubesheet groove weld leg :	1.333	3.000	Ok

Min Shell length of thk, (14.000) adj. to tubesheet: 219.275 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.014 mm
 Due to Pressure + Thermal Shell Compresses by : -0.014 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2778.93	1.000	9389.85	1.000
Tubesheet Shear Stress	11912.04	1.000	13077.69	1.000
Tube Tensile Stress	6650.76	1.000	17480.85	1.000
Tube Compressive Stress	9164.37	0.635	14054.42	1.000
Tube-Tubesheet Joint load	6650.76	1.000	17480.86	1.000
Shell Stress (Axial, Junction)	2778.93	1.000	3991.55	1.000
Tube Pressure Stress	16398.62	1.000	8894.20	1.000
Tubesheet Extension Stress	2660.46	...	No Calc	No Calc
Minimum MAWP	2660.46		3991.55	



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FileName : 192-C_BEM_Dia

Page 212 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9,2026

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	3339.07	0.998	9712.76	1.000
Tubesheet Shear Stress	17909.14	1.000	15412.01	1.000
Tube Tensile Stress	7915.63	1.000	20340.55	1.000
Tube Compressive Stress	13047.82	1.000	18303.24	1.000
Tube-Tubesheet Joint load	7915.63	1.000	20340.54	1.000
Shell Stress (Axial, Junction)	3339.07	0.998	6344.04	1.000
Tube Pressure Stress	16443.07	1.000	10612.12	1.000
Tubesheet Extension Stress	2660.46	...	No Calc	No Calc
Minimum MAPnc	2660.46		6344.04	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0101	0.0101	...	...	...	...
2	-0.0137	-0.0138	...	...	...	...
3	-0.0036	-0.0037	...	...	...	...
4	...	...	...	...	...	...
5	0.0101	0.0101	...	...	...	...
6	-0.0137	-0.0138	...	...	...	...
7	-0.0036	-0.0037	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(80/4, min(80, 14))
= 20.000 mm

Thickness Ratio = 0.34, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
= tubesheet thickness/4
= 80/4
= 20.000 mm

Thickness Ratio = 0.34, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(900/2660, 520/6344)
= 0.338



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FileName : 192-C_BEM_Dia

Page 213 of 214

ASME TS Calc: DESIGN 1

Case: 1 10:11a Sep 9, 2026

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FileName : 192-C_BEM_Dia

Page 214 of 214

Problems/Failures Summary:

Step: 34 10:11am Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: